



FROM RECOVERY TO RESILIENCE: LOCALIZING NATIONAL POLICIES AND FRAMEWORKS FOR MULTI-HAZARD RESILIENCE"



NEPAL SHELTER FORUM 2024



Government of Nepal
Ministry of Home Affairs
National Disaster Risk Reduction and Management Authority
Singhadurbar, Kathmandu

PROCEEDING REPORT







Acknowledgement

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Abbreviations

ASF	Asia Pacific Shelter Forum
BEEN	Building Energy Efficiency in Nepal
CBDRR	Community based Disaster Risk reduction
CDO	Chief District Officer
COP	Conference of Parties
CLPIU	Central Level Project Implementation Unit
CRS	Catholic Relief Services
CSEB	Compressed Stabilized Earth Brick
CSOs	Civil Society Organizations
DAO	District Administration Office
DDA	Detailed Damage Assessment
DDMC	District Disaster Management Committee
DHM	Department of Hydrology and Meteorology
DHSUD	Department of Human Settlements and Urban Development
DMG	Department of Mines and Geology
DPOs	Development Partner Organization
DRR	Disaster Risk Reduction
DRRM	Disaster Risk Reduction and Management
DUDBC	Department of Urban Development and Building Construction
DWRI	Department of Water Resources and Irrigation
EWS	Early Warning System
FCDO	Federation of Commonwealth Development Office
FECOFUN	Federation of Community Forests Users Nepal
FGDs	Focused Group Discussions
GDP	Gross Development Product
GEDSI	Gender Equality Disability and Social Inclusion
GoN	Government of Nepal
GRID	Green Resilient Inclusive Development
GRID SAP	Strategic Action Plan
HfH	Habitat for Humanity Nepal
HVRCA	Hazard, Vulnerability, Risk and Criticality Assessment
I/NGOs	International and/National Non-Governmental Organizations

ICT	Information Communication and technology
IFRC	International Federation of Red Cross and Red Crescent Societies
IOE	Institute of Engineering
IOM	International Organization of Migration
JICA	Japan International Cooperation Agency
LDMC	Local Disaster Management Committee
LNGOs	Local and National NGOs
MDTF	Multi Donors Trust Fund
MoEWRI	Ministry of Energy Water Resources and Irrigation
MoFAGA	Ministry of Federal Affairs and General Administration
MoFE	Ministry of Forests and Environment
MoHA	Ministry of Home Affairs
MoIAL	Ministry of Internal Affairs and Law
MoLCPA	Ministry of Land Cooperatives and Poverty Alleviation
MoUD	Ministry of Urban Development
MoWCSC	Ministry of Women, Children and Senior Citizens
NbS	Nature based Solutions
NARMIN	National Association of Rural Municipalities in Nepal
NDRF	National Disaster Resilience Framework
NDRRMA	National Disaster Risk Reduction and Management Authority
NFDN	National Federation of Disabled Nepal
NHSRP	National Housing and Settlement Resilience Platform
NPC	National Planning Commission
NPR/NRs	Nepali Rupees
NRA	National Reconstruction Authority
NRCS	Nepal Red Cross Society
NSET	National Society for Earthquake Technology Nepal
PDNA	Post Disaster Needs Assessment
RIMES	Regional Integrated Multi-Hazard Early Warning System for Africa and Asia
RMIS	Reconstruction MIS
RSLUP	Risk Sensitive Land Use Planning
SSA	Social Security Allowance
SYPanel	Say Yes Panel
ToR	Terms of Reference
UKAid	United Kingdom Agency for International Development
UNDP	United Nations Development Programme
UNOPS	United Nations Office for Project Services
USAID	United States Agency for International Development
USD	United States Dollars
WASH	Water Sanitation and Hygiene
WB	World Bank
WVI	World Vision International



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Executive Summary

The 2nd Nepal Shelter Forum (NSF 2024), a two-days national conference, was held in Kathmandu from May 7th to 8th, 2024 with the theme “From Recovery to Resilience: Localizing National Policies and Frameworks for Multi-Hazard Resilience” and was organized by the National Disaster Risk Reduction and Management Authority (NDRRMA). The event aimed to discuss policies, frameworks, and solutions to enhance disaster resilience at all levels of government and within communities. The conference was inaugurated by the Chief Guest and Honorable Mr. Bhanubhakta Joshi, Federal Affairs and General Administration Minister (MoFAGA), and there was special presence by the Honorable Chief Minister of Karnali Province, Mr. Yamlal Kandel. During the inauguration session, Mr. Rameshwor Marhatta, Joint Secretary of NDRRMA, welcomed the participants by giving the opening session remarks. Ms. Reshma Shrestha, Deputy National Coordinator of the NHSRP gracefully welcomed the participants by sharing the objectives of the event. Mr. Usec Henry Yap from the Philippines Department of Human Settlements and Urban Development

(DHSUD) also delivered a virtual welcome message as an organizer of the Asia Pacific Shelter and Settlements Forum (ASF 2024).

During the closing session of NSF 2024, Deputy Prime Minister and Home Minister Mr. Rabi Lamichhane announced the approval of “Disaster Affected Private Housing Retrofitting, Reconstruction, and Resettlement Tranche Disbursement Procedure 2081” with amendments. Minister Lamichhane reaffirmed the government’s commitment to ensuring the right to housing of disaster affected people as guaranteed by the Constitution of Nepal.

NSF2024 featured a total of nine panel sessions, which contributed to a better understanding of sector-specific needs and integrated approaches to multi-hazard disaster resilience and recovery. The discussions provided useful suggestions for future recovery efforts, focusing on Gender Equality, Disability and Social Inclusion (GEDSI), Green Resilient and Inclusive Development (GRID), and the use of Information Communication and Technology (ICT) in disaster preparedness and recovery works.

On the first day of the program, there was a lot of discussion about stocktaking of 2015 Nepal earthquake reconstruction, by connecting with the forthcoming recovery works related to the earthquake in West Nepal that devastated many parts of Karnali and Sudurpaschim province. A policy dialogue was held to reflect on the Jajarkot earthquake and find ways to improve multi-hazard disaster response and management, as well as to enhance approaches and technologies for resilient shelter.

On the second day of the program, a panel session discussed rampantly on an integrated approach to reconstruction and rehabilitation and also on emphasizing the use of ICT for disaster preparedness, response, and recovery. Similarly, discussions also focused on promoting inclusive resilience by integrating GEDSI and advocating for GRID principles, along with the coverage of impact of climate change on housing and settlements and explored innovative practices for sustainable housing and settlements.

During the presentation session, experts highlighted the key milestones in Nepal's reconstruction efforts since 2015, and reflected upon how the relevant DRRM efforts, appropriate policies and governance improvements can enhance disaster response and recovery which was linked with ongoing efforts in the Karnali region. The sharing of collaborative efforts of development partners and organizations in response to the West Nepal earthquake highlighted how the early response and relief works, along with socio-technical assistance supported vulnerable people in transitioning from emergency to transitional shelter, as well as restoring their livelihoods and dignity. The modality of Start Fund Nepal and the contingency pooled fund managed by NGOs to provide rapid funding for small to medium-scale crises in Nepal was also shared. The ongoing initiatives of Building Energy Efficiency in Nepal (BEEN) Project, shed insights on the importance of energy efficiency in shelter design and construction, particularly in post-disaster recovery and reconstruction

The event was attended by distinguished guests, including government officials, development partners, humanitarian agencies, researchers, and media personnel. Notable personalities such as Members of the National Planning Commission - Dr. Araniko Pande and Ms. Sangita Singh, along

with Former Urban Development Minister Ms. Ramkumari Jhankri, Former Foreign Minister Dr. Bimala Rai Poudyal, Former Chief Executive Officer of NRA Mr. Sushil Gyewali, along with Mayors from various municipalities, and Chairpersons of rural municipalities marked the presence as the panelists. The opening session had over 300 participants, including government officials from multiple ministries and departments, sectoral line agencies, DRRM experts and specialists from bilateral donor agencies, I/NGOs, humanitarian organizations, academics, scholars, students, security agencies, and media personnel. Their presence and active involvement set an energetic tone for the event.

NHSRP, as the NSF Secretariat, managed the coordination, operations, and overall management of NSF2024 program activities. The financial contributors to NSF2024 were the World Bank Group/MDTF (Multi-Donor Trust Fund), UNDP, and 13 development organizations of Nepal including UNDP, IOM, CRS Nepal, Habitat for Humanity Nepal (HfH), Nepal Red Cross Society/International Federation of Red Cross and Red Crescent Societies (IFRC), Save the Children, World Vision International, Caritas Nepal, BEEN Co-funded by European Union, Build up Nepal, Say Yes Panel (SY Panel) and NHSRP.

NSF was conceived as a national platform through discussions at the Asia Shelter Forum (ASF) International Conference held in November 2021 in Kathmandu, Nepal. It was agreed among all relevant governmental and non-governmental shelter, settlements, and DRRM practitioners to hold an annual national event like NSF and work towards institutionalizing engagement by building on existing coordination mechanisms to localize best practices in shelter, housing, and settlements. Building on the insights gained from the first edition of national forum held in Kathmandu in April 2023 around the Memorial Day of 2015 Nepal earthquake, the NSF 2024 aimed to foster collaborative efforts and innovative approaches to ensure safe and dignified shelter and housing for all Nepalis, resilient in the face of climate and environmental changes.

Background

This proceeding report summarizes the happenings of 2nd NSF 2024, highlighting the discussions held in sessions and consolidating the insights shared by experts and guests. The NSF2024 took place on May 7th and 8th, 2024 in Kathmandu, with the theme of “From Recovery to Resilience: Localizing National Policies and Frameworks for Multi Hazard Resilience.” The conference, held in leadership and guidance by the NDRRMA, was a successful two-day event aimed at promoting collaboration for multi-hazard resilience by taking stock of lessons learned from 2015 Nepal earthquake reconstruction and recovery programs. The discussions at NSF 2024 covered integrated approaches to reconstruction and recovery, localized solutions for fire risk preparedness, GRID, GEDSI, ICT, and climate-induced loss and damage. The roles of government bodies and partner organizations in effectively responding to disasters of varying extent and scale were also a key highlights in several sessions.



- Bring together stakeholders from Nepal's shelter, housing, and settlement sectors to collaboratively discuss, devise solutions, and provide a platform for government bodies, development partners and organizations, private sectors, civil society, and diverse stakeholders to share insights, promote best practices, and collectively address emerging challenges.
- Advocate for and disseminate National Policies, Frameworks, and Procedures related to the

1.1 Objectives

The main goal of NSF 2024 was to address policies, frameworks, and solutions to improve disaster resilience at all levels of government and within communities. The discussions in the NSF event sought to promote collaboration for multi-hazard resilience by leveraging innovations, sharing promising solutions, and fostering greater connectedness and collaboration among stakeholders for improving the resilience of communities at risk. The event largely aimed to-





Shelter and Housing Sector.

- Focus on learnings from the recent earthquake response in West Nepal as a case study to facilitate discussions around the Government of Nepal and partner organizations' plans for sustainable and inclusive development through the promotion of green and local construction technologies.
- Foster collaboration between organizations and private enterprises with the capacity to impact and scale up Nepal's low-cost, affordable housing sector with sustainable construction technologies.
- Explore best practices and innovative approaches in disaster risk preparedness, needed in communities to ensure the provision of safe, adequate, and dignified shelters, housing, and settlements, with proper access to land and finances as cross cutting themes.
- Integrate the knowledge and reflections gained from the NSF into the agenda of the Asia Pacific Shelter Forum (ASF) 2024 to foster broader dissemination, discussion, and international sharing of insights.

Official inauguration of NSF 2024 was done by the Hon. Minister of Federal Affairs and General Administration, in presence of multitude of different stakeholders of government (Federal, Provincial, and Local), development representatives from I/ NGOs, civil societies, academics, private sector stakeholders, activists, students and media at large. These diverse stakeholders assembled to discuss and share the perspectives through several panel sessions that encompassed integrated approaches to reconstruction and recovery, multi-hazard disaster



recovery, fire disaster preparedness measures as well as localized and innovative solutions to GRID, GEDSI, ICT, climate induced loss and damage along with identifying the coordination and collaboration support from all concerned sectors.

1.2 Participants

A total of 318 participants (235 male and 83 female) attended two-days national conference of NSF 2024. Of the total attendees, 74% male and 26% female participants attended the conference.

1.3 Thematic Outline

The national conference of 2nd NSF featured nine parallel sessions and two presentation sessions; each encircled around on the main theme "From Recovery to Resilience: Localizing Policies and



Frameworks for Multi Hazard Resilience". The panel sessions had the presence of several distinguished experts in the fields of multi-hazard disaster recovery, resilience, governance, post-disaster housing reconstruction, climate change adaptation, gender and social inclusion and greening the shelter approaches. The presentation session showcased the reflection of post-disaster initiatives of Central Level Project Implementation Unit/ Ministry of Urban Development (CLPIU/MoUD), the humanitarian early responses and temporary shelter recovery works done by the development organizations, along with the compiled initiatives of Start Fund Network. All the sessions comprised deep interactions followed by question and answers and the keynote remarks from the session chairs.

The sub-thematic program outline for the NSF event has been developed to address the current challenges and needs of the sector which is as follows –

- Localization of Recovery and Reconstruction of Shelter and Settlement
- GRID and Affordable Practices
- Integrated Reconstruction and Recovery
- ICT Engagement in DRRM
- GEDSI in Recovery and Reconstruction
- Climate Change and Loss and Damage





2

Welcome and Inauguration

On the first day of NSF 2024 National Conference, Ms. Ranjana Shrestha, Communication Outreach Expert of NDRRMA led the Master of Ceremony of the opening session and graciously welcomed all the invited chief guests, special guests and dignitaries representing governmental agencies of Nepal, distinguished participants from donors organizations, participants from I/NGOs, private sectors, academics, media personnel, staff members of NSF Secretariat and NSF Organizing Committee at large for their participation in second edition of NSF 2024. The conference



was inaugurated by the Honorable Minister of MoFAGA, Mr. Bhanubhakta Joshi by lighting the traditional lamp as per Nepali traditions. All the participants stood from their respective seats to recite the National Anthem. Similarly, a moment of silence for 1 minute was also observed to mark remembrance to all the lives lost by the big disasters such as 2015 Nepal earthquake and others.



2.1 Opening remarks by Joint Secretary of NDRRMA Mr. Rameshwor Marahattha



Joint Secretary of NDRRMA, Mr. Rameshwor Marahattha, welcomed the participants at the 2nd NSF 2024 by highlighting the constitutional guarantee of adequate housing measures for disaster affected people. A pre-earthquake survey revealed only 5% of school buildings were in good condition, a situation worsened by the 2015 Nepal earthquake, with similar issues expected for residential private houses. He discussed the post-earthquake response in West Nepal, noting the need for Detailed Damage Assessment (DDA) and the reconstruction of over 100,000 houses. The Preliminary Damage and Needs Assessment (PDNA) report estimated a need for NPR 38 arba (USD 284.7 million) for private housing reconstruction that identified key approaches including owner-driven reconstruction, build back better, and using green local materials, along with promotion of partnerships approach by local bodies with partner organizations. Mr. Marahattha also outlined the new procedure for disaster-affected housing retrofitting and reconstruction, emphasizing local accountability and collaboration for improved housing resilience in Nepal.

Youtube of opening session: <https://bit.ly/4cWoX87>



2.2 Opening Remarks from Deputy National Coordinator of NHSRP, Ms. Reshma Shrestha

Ms. Shrestha expressed warm gratitude and gave the opening welcome remarks on behalf of the Organizing Committee of the 2nd Nepal Shelter Forum 2024 and its Secretariat. She informed the participants that the Nepal Shelter Forum is the national version of the Asia Pacific Shelter Forum (ASF). 2nd NSF conference followed the trend of last year's Nepal Shelter Forum, which focused on implementing Nepal's Green, Resilient, and Inclusive Development Pathway and held in Kathmandu from April 25-26, 2023, that resulted in commitments to improve coordination between government tiers for disaster preparedness, risk reduction, and resilient recovery. The 2nd NSF aims to enhance cooperation and contribute to learning exchange between government, donors



and development organizations involved in disaster risk reduction and housing resilience. Ms. Shrestha welcomed participants, emphasizing the goal of achieving multi-resource sustainability through green solutions and gender inclusion. She hoped the discussions would document lesson learned and actionable strategies focused on localization, inclusiveness, and resilience, urging active participation from everyone.

Youtube of opening session: <https://bit.ly/4cWoX87>



2.3 Message from the Philippines DHSUD Organizer for the Asia Pacific Shelter Forum (ASF 2024), Mr. Usec Henry Yap

Mr. Yap congratulated the Nepal Government, the participating organizations, and the NSF 2024 Secretariat for their remarkable achievement in creating a shared platform. He emphasized the distinct characteristics and challenges of urban and rural settings, underscoring the need for diverse shelter policies tailored to these unique contexts. He shed lights that platforms like Asia Shelter Forums are crucial for fostering partnerships and collaborations among like-minded forces and stressed organizers on taking forward the consolidating and revitalizing the indigenous knowledge in DRRM efforts while ensuring inclusivity, sustainability, and resilience as essential factors for effective disaster response and shelter policy development. Using NSF 2024 platform, Mr. Yap also invited the interested personnels and organizations to attend the ASF 2024 to be held in Manila, Philippines.

2.4 Remarks from Chief Minister of Karnali Province, Mr. Yamlal Kandel

Chief Minister Kandel emphasized the urgent need to construct houses for Jajarkot earthquake survivors, expressing concern over affected families still living in makeshift tents as the monsoon season approaches. Speaking at the NSF inauguration,

he noted that relief for building shelters has not yet been provided. Federal, provincial, and local governments are collaborating to initiate shelter



construction. Kandel highlighted the increasing frequency of disasters, such as forest fires, floods, and landslides, and questioned the capacities of earthquake affected people to endure the monsoon rains in torn tents. He criticized existing laws and regulations for being inadequate in disaster management and noted that some of the affected people have not received the promised Rs 50,000 for shelters. He stressed the importance of effectively implementing the disaster response action plan, pointing out that Karnali Province has a disaster-caused death rate of five people per 100,000 population. Mr. Kandel called on all levels of government and stakeholders to improve disaster

response conditions in Karnali and highlighted the need for appropriate preparation against potential future disasters that could affect 170,000 people and more.

2.5 Opening Session Remarks by MoFAGA Minister, Mr. Bhanubhakta Joshi

Minister Joshi emphasized that Nepal faces significant natural disasters like lightning, fires, earthquakes, and monsoons, and noted progress in addressing multiple hazards beyond just earthquakes. He



stressed the importance of coordination between federal, provincial, and local governments for effective reconstruction, highlighting the lack of financial resources at the local level as a barrier in efficient disaster response. Minister Joshi called for stronger laws, policies, and regulations to ensure smooth reconstruction processes and addressed the challenges in implementing current policies during calamities such as landslides, floods, and fires. He emphasized the need for clarity and proper execution of authority across all government levels. Discussing disaster-hit areas like Jajarkot and Rukum West, Joshi stressed the urgency of providing shelter with the monsoon approaching and urged all levels of government and supporting organizations to be proactive. He underscored the critical role of local governments in early disaster response, as they are the first responders. Minister Joshi expected that discussions in forums like NSF aimed to find ways to facilitate work in affected areas, implement lessons from past earthquakes, and promote the “build back better” approach.

2.6 Closing Remarks of the Opening Session Mr. Anil Pokhrel, Chief Executive of NDRRMA

Mr. Pokhrel reminisces the lesson learned from 2015 Nepal earthquake reconstruction by highlighting that the post-earthquake reconstruction guideline provided a roadmap for future disasters and stressed the need for a comprehensive approach to both minor and major disasters. Pokhrel shed light on the limited capacity of local levels to



assist disaster-affected individuals due to financial, social, and technical constraints, especially during major disasters. He advocated for creating local employment opportunities to revive the economy and reduce migration contextualizing the major disaster recovery works in affected regions. Regarding the Jajarkot earthquake, he noted that DDA and local data verification are pending and challenging that are notably the crucial steps for planning the reconstruction ahead. He also pointed out Nepal’s inadequate emergency services and firefighting capabilities, stressing the need for volunteers training in fire response. Key takeaways of Pokhrel’s remarks include the importance of an integrated approach to disaster risk management, ensuring no one is left behind, and addressing both major and minor disasters equally. He underscored the significance of creating livelihood opportunities during disasters and stressed the need for coordination among stakeholders to collaborate on innovative solutions, and resources sharing for effective disaster response, reconstruction, and recovery.

Youtube: <https://bit.ly/4cWoX87>

Presentation Session

3.1 Reflection of West Nepal Earthquake Sharing on Government Perspective from Mr. Koshnath Adhikari, Project Director, CLPIU/ MoUD

Mr. Adhikari highlighted the key achievements from the 2015 Nepal earthquake reconstruction efforts, noting that approximately 80.99% of over 800,000 earthquake damaged houses have been rebuilt. The reconstruction phase prioritized private housing, followed by public infrastructure for health and education, government buildings, and other structures. Despite efforts to include all affected families, nearly 50,000 households did not seek housing support. Among those who did participate, 64,783 families received the first installment but not the second; and 27,563 families received the first two installments but not the third, due to a lack of financing options to complete their homes.

Responsibility for continued housing reconstruction has shifted from NDRRMA to MoUD, but funding shortages have hindered the progress. Mr. Adhikari noted that the CLPIU implemented the Integrated Settlements Project in two ways: i) for displaced and relocated individuals, and ii) for those who migrated to urban areas. Regarding the Jajarkot earthquake response, Mr. Adhikari emphasized the need for integrated livelihood opportunities of the affected families within existing settlements through better planning. Similarly, he stressed on proper information flow and communication with disaster affected families about timely submission of reconstruction tranche disbursement form, public sensitization on adherence to building codes and safe shelter construction as the important measures of post-disaster recovery approach.

Presentation: <https://bit.ly/4cZwJOq>

Youtube: <https://bit.ly/3X6tILO>



3.2 Sharing the Nepal Shelter Cluster Initiatives by Mr Manish Timsina, Shelter Coordinator, NRCS



Mr. Timsina drew attention of participants by sharing that the MoUD leads the Shelter Cluster in Nepal, with support from IFRC/NRCS. Following the Jajarkot earthquake, various organizations have collaborated to provide emergency and transitional shelter assistance mainly with an aim to support vulnerable people to move from emergency to transitional shelter, restoring regular livelihoods and dignity. Organizations such as NRCS, Save the Children, HfH, World Vision International, CARE Nepal, Plan International, Caritas Nepal, CRS Nepal, IFRC along with donor partners like IOM, UNICEF, WFP and UNDP were thoroughly involved in lending the disaster response along with technical support and guidance for temporary shelter construction too. Government coordination mechanism body, NDRRMA approved and endorsed the temporary shelter construction guideline 2023, which paved a promising way forward for partner organizations in offering social technical assistance to the earthquake affected families in temporary shelter construction, in close coordination with DDMC and local municipalities. Use of information boards in public places and information dissemination through local radio channels and telecommunication messaging were rampantly utilized on the temporary shelter guideline and government grant support. Help

desk support services were provided by NRCS volunteers in local municipalities to disseminate transitional shelter guidelines, along with expediting the registration of beneficiaries, and managing installment release processes.

Presentation: <https://bit.ly/3MTo9X7>

Youtube: <https://bit.ly/3XgW1Sa>

3.3 Sharing on Start Fund Network by Mr. Nirajan Pokhrel, Project Manager, Rural Reconstruction Nepal

Funded by the UKAid's Foreign, Commonwealth and Development Office (FCDO), this member-led funding mechanism has been designed and implemented by local, national, and international organizations to support the most vulnerable crises affected and at-risk affected communities. This national fund is promoting localization in response and crisis anticipation from the very beginning with effective representation of Local and National NGOs (LNNGOs) in decision-making roles and channeling most of its funds to these organizations



through INGOs. Modeled on the Global Start Fund mechanism, Start Fund Nepal ensures rapid and anticipatory funding to the affected people through its swift decision-making (within 72 hours) and its programmatic reach on the ground within 7 days and project phase lasts for a period of 60 days. Representing Start Fund Network, Mr. Pokhrel informed the participants about the the initiatives implemented by these organizations with coverage on owner-driven approaches to transitional shelters, cash-for-work programs for shelter construction, toilets, semi-permanent schools, and mason training. These initiatives have been carried out following various disasters in Nepal, such as the Doti, Bajura and Jajarkot earthquakes, Rautahat fires, Gorkha landslides, and Taplejung floods.

Presentation: <https://bit.ly/3z9xeaS>

Youtube: <https://bit.ly/3z4gaDa>

3.4 Sharing on BUILDING Energy Efficient Shelter in Nepal by Dr. Daniel Neyer, Project Leader, Building Energy Efficiency in Nepal (BEEN)

The BEEN project, funded by the EU under the SWITCH-Asia Programme and led by the University of Innsbruck (Austria) aims to promote low carbon

and resource-efficient construction in Nepal's building sector from 2022 to 2026. Implemented by MinErgy, the project facilitates climate-friendly building design, construction, and renovation. Dr. Neyer highlighted key aspects including training on passive designs, building materials, energy-efficient systems, and renewable energy. BEEN also produces manuals, guidelines, and baseline reports to promote energy efficiency in construction practices particularly in post-disaster contexts. Dr. Neyer stressed the importance of thermal comfort and climate-responsive design. He recommended maintaining indoor temperatures between 18°C and 32°C for safety and comfort, underscoring the influence of building materials and design on internal temperatures. The use of passive design strategies like orientation and suitable materials was advocated for effective climate management. He emphasized designing shelters according to local climatic conditions to avoid a one-size-fits-all approach, adapting designs to meet regional needs and cultural preferences. His presentation outlined a holistic approach to energy-efficient shelter design, aiming for safer and sustainable living conditions across diverse climates.

Youtube: [Youtube: https://bit.ly/3zaOWe3](https://bit.ly/3zaOWe3)

Presentation: <https://bit.ly/47hmaow>



4

Panel Session



Mr. Rijan Gajurel and Ms. Preeti Karki represented NSF Secretariat and served as the Master of Ceremonies for panel sessions at the 2nd Nepal Shelter Forum. Setting grounds for panel sessions, Mr. Gajurel provided crucial background information, focusing on the impact of recent earthquakes in Doti, Bajhang, and Jajarkot, which collectively damaged 95,787 households according to NDRRMA's Bipad

Portal data. Additionally, floods, landslides, and fires affected a total of 18,755,610, and 34,347 households respectively, causing complete or partial damage. Mr. Gajurel also highlighted the humanitarian early response efforts from various sectors and institutions following these disasters.

Presentation: <https://bit.ly/4dQaA6u>





4.1 Panel Session: Multi-hazard Recovery and Reconstruction Governance

Session Facilitator: Mr. Jagdish Kharel, Senior Journalist, Image Channel Television

This policy dialogue panel session was purposefully designed to raise discussions to evaluate the governance framework related to multi-hazard recovery and reconstruction efforts in Nepal. To unpack the policy dialogue, the Session Facilitator Mr. Jagdish raised questions about the barriers to conducting a DDA following the West Nepal earthquake. The Facilitator also asked about disaster recovery governance structures at federal, provincial, and local levels that impacts disaster management efforts, particularly regarding data management and identifying real beneficiaries at the local level. Panelists suggested identifying challenges and gaps and proposing actionable recommendations to the Government of Nepal to enhance policy implementation and improve disaster response and resilience. The panelists shed light on these topics, addressing disaster management fund policy structures, land issues, lessons learned from the 2015 post-earthquake reconstruction, and the application of these lessons in integrated reconstruction efforts.



Panelist: Mr. Yamlal Kandel, Chief Minister, Karnali Province



Minister Kandel emphasized the urgent need to disseminate the PDNA report and called for execution of DDA with proper financial and human resources planning. He urged the NDRRMA to speed up the verification and disbursement of funds for temporary shelter construction. Kandel highlighted the critical role of municipal authorities in disaster response but noted challenges in data verification and funding. He advocated for revising emergency response policies to clarify financial roles and responsibilities. He also discussed the land issues and insurance support in Karnali, stressing the impact of political influence. Using the platform Nepal Shelter Forum, he called for robust, adaptive policy frameworks to ensure resilient recovery and reconstruction, proper use of disaster management funds, and housing solutions that blend cultural and modern elements to reduce disaster risk.

Panelist: Ms. Ramkumari Jhakri, Former Minister of Ministry of Urban Development (MoUD)

Former Minister Jhakri highlighted significant gaps in Nepal's disaster management and governance, particularly the unclear mechanism for utilizing disaster funds, leading to inadequate financial resources for disaster management. She emphasized on utilizing the learning from the 2015 Nepal earthquake to improve policies and expedite response and reconstruction efforts post-Jajarkot earthquake. Administrative inefficiencies



and bureaucratic delays hinders the reconstruction process. Given Nepal's vulnerability to various natural disasters, Jhakri stressed the need for robust, adaptive strategies for response, recovery, and reconstruction. She noted the critical issue of non-compliance with building codes, making many government buildings vulnerable. Jhakri called for a comprehensive emergency response policy, effective funding management, and urged NDRRMA's Chief Executive to advocate for effective recovery efforts with the Prime Minister.

Panelist: Mr. Sushil Gyewali, Former Chief Executive, National Reconstruction Authority



Mr. Gyewali highlighted key achievements and lessons from the 2015 Gorkha earthquake recovery programs, emphasizing on the need for rapid response mechanisms, community involvement, technical capacity building, and owner-driven resilient building practices. He stressed the importance of mandatory and strict enforcement of building regulations to enhance structural resilience. Gyewali recommended revisiting and revising disaster management strategies rather than starting anew, advocating for a clear action plan from NDRRMA based on the PDNA report. He called for finance-based prioritization, result-based actions, decentralization of authority, and clear delineation of roles between NDRRMA and central authorities. Gyewali emphasized the necessity of partnership and coordination for effective disaster management, advocating for efficient mobilization and allocation of resources to support recovery efforts.

Panelist: Mr. Anil Pokhrel, Chief Executive, NDRRMA



Mr. Pokhrel discussed critical issues in Nepal's disaster management and reconstruction efforts by noting that the NDRRMA's limited budget of NPR 3 arba (USD 22.4 million) restricted efficient emergency activities and services along with safer reconstruction of private housings. Funding shortages have delayed the DDA along with planning of long-term housing reconstruction in Karnali,

compounded by delays in finalizing the PDNA report. He emphasized the need for accurate databases from local level to expedite response efforts and disbursement of the second installment for shelter construction. He advocated for integrating multi-hazard approaches in recovery strategies. In his concluding remarks, Pokhrel highlighted the need for strict building code enforcement, referencing a 60-point recommendation submission to the Prime Minister from NDRRMA's side. Mr. Pokhrel emphasized the importance of collaboration among stakeholders and called for transparent and accountable mechanisms to ensure resources reach intended beneficiaries, stressing the need for sustainable and resilient infrastructure.

Key takeaway message of the session

- NDRRMA requires a comprehensive action plan, utilizing the PDNA, to prioritize and finalize actions based on financial considerations.
- Delegate authority to the provincial level to ensure result-oriented responses and enhance disaster management effectiveness.
- Incorporate the lessons learned from the 2015 Nepal earthquake to expedite response and reconstruction efforts and improve policies.
- Create adaptive and inclusive policies to facilitate systematic, efficient, and resilient recovery and reconstruction, accounting for the interrelatedness of different hazards.
- Strengthen resilience and safety by rigorously enforcing building codes in both the public and private sectors.

Youtube: <https://bit.ly/3TifzVh>





4.2 Panel Session: Reflection of West Nepal Earthquake

Session Facilitator: Mr. Jagdish Kharel,
Senior Journalist, Image Channel
Television

Session Chair, Mr. Sushil Baidhya,
Secretary, MoIAL, Sudurpaschim Province



This panel session mainly reflected on the response and recovery mechanisms for areas affected by the Jajarkot Earthquake. Session Facilitator Mr. Jagdish raised questions on key relief and early response efforts by the government and various agencies (NRCS, UNDP, I/NGOs, Start Fund Network and others), examining the current situation, successes, gaps, challenges, and future strategies for recovery, reconstruction, and resilience in Western Nepal. Facilitator unpacked the discussions on the implementation of temporary shelter guidelines, the Management Information System (MIS) for

tracking earthquake beneficiaries and grant disbursements, challenges during the response phase, and the development of multi-hazard guidelines. The discussion also covered lessons learned from the Jajarkot earthquake response, priority reconstruction areas at the Palika level, plans for geohazard assessment, and settlement relocation. Facilitator further explored the integration of national and international efforts to enhance disaster preparedness and resilience-shelter solutions.

**Panelist: Dr. Dijan Bhattarai, Under
Secretary and Spokesperson, NDRRMA**



Dr. Bhattarai highlighted the severe challenges faced by people affected by the 2023 Jajarkot earthquake in Western Nepal. He described the ongoing difficulties for families enduring extreme

cold weather and upcoming monsoon without permanent housing, increasing their vulnerability. The earthquake's impact in Doti, Bajura, Bajhang, and Jajarkot resulted in significant losses of lives and homes, with approximately 86,000 families identified as needing support. Of these, 76,875 beneficiaries have signed agreements for temporary shelters, and 71,378 shelters have been constructed, but many families still await the second tranche of funding, causing a fund balance in municipalities. Dr. Bhattarai added that a total of NPR 4.75 billion (USD 355.9 million) is required, with NPR 3.16 billion (USD 23.6 million) already sent to the District Administration Offices of Jajarkot, Rukum West, Salyan, Bajhang, Bajura, and Jumla. NPR 2.07 billion (USD 20.2 million) has been transferred to municipalities, and NPR 1.92 billion (USD 14.3 million) to beneficiaries. Dr. Bhattarai informed the audiences on announcement of the recent approval of the "Disaster Affected Private Housing Retrofitting, Reconstruction, and Resettlement Tranche Disbursement Procedure 2081" on April 15, 2024, outlining a three-stage funding process for beneficiaries. The next crucial step is completing the DDA for which, NDRRMA has requested NPR 200 million (USD 1.4 million) from the Ministry of Finance.

Panelist: Mr. Chandra Prakash Gharti
Magar, Mayor, Bheri Municipality, Jajarkot district



Mayor Magar discussed the impact of the November 2023 Jajarkot earthquake on Bheri Municipality, highlighting that all affected beneficiaries have been identified. Currently 6,631 families have received the first installment of support, but delays in receiving the second installment stem from uncertainties over whether their old houses must be demolished to qualify for further assistance. He expressed concerns over mixed advice from different stakeholders, complicating implementation, along with unclear roles among government departments involved in reconstruction. Mayor Magar stressed the need for solution-oriented policies aligned with local realities. He advocated revising policies that mandate house demolition for eligibility, emphasizing that not all homes require demolition. Furthermore, he called for streamlining the NDRRMA's Bipad portal to ensure efficient reconstruction progress and proposed a transparent policy framework responsive to the actual needs of affected families, aiming for more effective and timely reconstruction efforts.

Panelist: Mr. Umesh Prasad Dhakal,
Executive Director, NRCS



Mr. Dhakal emphasized NRCS's role in search, rescue, and relief during the initial response to disasters. As the relief phase transitions to recovery, he stressed the urgency of implementing a comprehensive recovery plan by December 2024 for systematic reconstruction in earthquake-affected areas of West Nepal. He highlighted the prolonged nature

of the reconstruction phase and its vulnerability to upcoming monsoons, underscoring the need for expedited distribution of the second installment of aid and faster construction of transitional shelters. He urged development partners to mobilize resources to support these efforts. While the recent approval of reconstruction guidelines is a milestone, Mr. Dhakal emphasized the critical importance of completing the DDA to advance reconstruction. He expressed concern over communities losing hope due to delays in support, emphasizing the importance of modest yet essential ground-level initiatives like software activities in the overall recovery process.

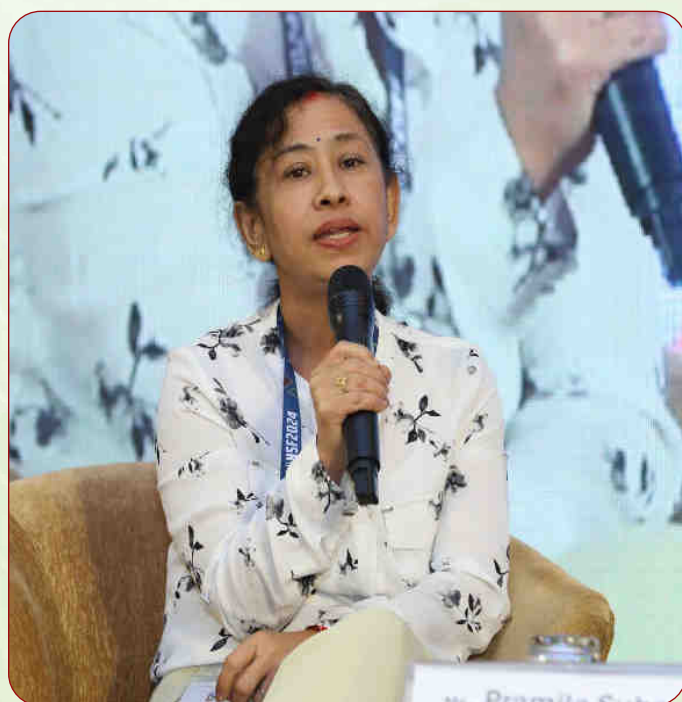
Panelist: Mr. Vijaya Singh, Assistant Resident Representative and Policy Advisor, UNDP Nepal



Mr. Singh emphasized UNDP's proactive role in early recovery initiatives. While UNDP does not directly oversee recovery and reconstruction efforts, it collaborates closely with Development Partner Organizations (DPOs) to address these critical needs. Singh highlighted the importance of fostering robust cooperation with the Government of Nepal at all levels—local, provincial, and central. This approach ensures that recovery efforts are well-aligned with national priorities and are

enriched by local knowledge and resources. Mr. Singh announced an upcoming UNDP socio-technical support program specifically designed for earthquake-prone areas. This program aims to provide holistic assistance by integrating social interventions with technical expertise to bolster resilience and preparedness. By partnering with government entities and leveraging the capabilities of DPOs, UNDP aspires to establish sustainable recovery frameworks that empower communities, encourage inclusive development, and reduce the impact of future earthquakes.

Panelist: Ms. Pramila Subedi, Program Manager, Start Fund Network



Ms. Subedi emphasized the collaborative achievements of multiple organizations working in West Nepal, as highlighted in the earlier presentation session of NSF. She noted that while the Start Fund primarily focuses on rapid response immediately after a disaster, development partners can play a crucial role in supporting the recovery and reconstruction phases in alignment with the Government's efforts. Ms. Subedi pointed out the existing confusion regarding whether old houses need to be demolished to receive the second tranche of funding, emphasizing the need for clarity on this issue. She highlighted the importance of close coordination with local governments and guidance from NDRRMA for effective implementation of development partners on the ground even for a short term basis. Additionally, Subedi highlighted

the challenge of duplicative efforts and stressed the need to minimize this duplication to ensure efficient and effective disaster response and recovery efforts.

**Concluding remarks from Session Chair,
Mr. Sushil Baidhya, Secretary, MoIAL,
Sudurpaschim Province**



Mr. Baidhya highlighted the ongoing challenges and efforts in Western Nepal and noted that Sudurpaschim has been significantly affected by two earthquakes in Doti and Bajhang districts. He underscored the urgent need for action on reconstruction by the state government. He informed that at the provincial level, Provincial Emergency Operation Centers (PEOCs) are actively involved in data management in the DRRM sector that also highlights streamlining data entry processes in Bipad Portal. Following the approval of the “Procedure on Disaster Affected Private Housing Retrofitting, Reconstruction, and Resettlement Tranche Disbursement 2081,” reconstruction activities are set to begin after the DDA, if conducted in a timely manner, bringing hope to the affected communities. Additionally, there is a growing emphasis on integrating DRR education into school curricula, indicating a proactive approach to building resilience from an early age. Mr. Baidhya praised the invaluable support provided by partner organizations in supplying essential human resources for DRR initiatives, underscoring the importance of collaborative efforts in effectively addressing regional challenges.

Key takeaways message of the session:

- The priority is to secure funds to conduct the DDA, as reconstruction is entirely dependent on the results of the DDA and the alignment of agency priorities.
- The genuine suffering of the families affected by the disaster must be prioritized and resolved through coordinated efforts in reconstruction as soon as possible.
- The guidelines for temporary shelters need to be clearly communicated, addressing the fact that the second tranche can be provided to families without demolishing their existing houses if they are safe.
- Prioritizing and allocating a budget for preparedness is key to ensuring that disasters, such as the earthquakes in the West, do not result in the level of damage and loss we currently observe.
- Skilled human resources are essential for the implementation of the reconstruction guidelines. From data entry to monitoring to capacity building, all initiatives require a significant number of human resources.

Youtube: <https://bit.ly/4gaP2TD>





4.3 Panel: Fire Risk and Preparedness Measures

Session Facilitator: Mr. Bhushan Tuladhar, Chief of Party, FHI 360



Session Chair: Mr. Machakaji Maharjan, Deputy Director General, DUDBC

The session aimed to raise awareness about household and wildfire risks, focusing on enhancing fire safety in shelter and settlement development. Session Facilitator Mr. Bhushan raised issues at local and implementation levels regarding fire risk, mitigation measures, and disaster response across three government tiers. Insights from recent seismic events were used to ignite the discussion by the facilitator in order to take constructive feedback on influence fire risk management and emergency response plans in Nepal, particularly in shelter and housing reconstruction. Panelists discussed

resource availability and local preparedness, linking these with federal-level efforts to increase public awareness through information mobilization. Facilitator also explored the need for building codes for fire recovery, alongside strategies to address wildfires in remote and ecologically sensitive areas, as well as in dense informal settlements by installation of fire alarms and development of evacuation plans. Provocative facilitation led the experts to discuss the best practices, suggest the adaptive measures including local and indigenous approaches to fire risk mitigation, in the context of climate change and air pollution challenges.



**Panelist: Mr. Baijnath Prasad Yadav,
Mayor, Madhav Narayan Municipality,
Rautahat district**



Mayor Yadav recounted the devastating fire in April 2024 that burned 21 houses in Bisunpurwa settlement, Madhav Narayan Municipality-4, Rautahat. Mayor Yadav recalled that fire is the major incident in several dense areas of Rautahat district in yesteryears too, though this year no human lives were lost by fire, however, destruction of cattle, food, clothing, and other belongings were damaged. In response, the Municipality initiated an awareness program stressing the importance of cooking within 9 AM in the morning. He highlighted poor electric installations in dense and thatch roof houses. Currently, the Rautahat district has only two fire brigades, causing delays in fire response. To address this, the Municipality is collaborating with other neighboring municipalities to enhance fire brigade availability. A survey of fire-affected households was conducted, and the data, integrated into the Reconstruction MIS with support from NHSRP, that awaits District Disaster Management Committee (DDMC) approval, after Local Disaster Management Committee (LDMC) endorsement. Mayor Yadav called for policy flexibility, stronger partnerships, and the support of DRRM stakeholders to improve fire risk management and preparedness at the local level.

**Panelist: Dr. Rejina Maskey Byanju,
Professor, Central Department of
Environmental Science, Tribhuvan
University**

Dr. Byanju highlighted the alarming rise in air pollution in Kathmandu linked to over 5,000 forest fire incidents during Nepal's dry season, contributing to elevated levels of CO₂, NO₂, and particulate matter. These pollutants severely impact health, the environment, and ecosystems. Current studies are insufficient, but a year-long deep dive study aims to provide more comprehensive data on fire incidents, their spread, causes, and mitigation measures. Dr. Byanju also stressed the need for enhanced research, awareness, and implementation of fire mitigation strategies. She advocated for technology solutions like air purifiers and converting forest waste into energy to control air pollution sustainably. She suggested adaptive management to combat air pollution by implementing an odd-even vehicle system, promoting indoor air purifiers, and advocating for masks to protect public health amidst increasing forest fire threats.



Panelist: Mr. Ranjan Dhungel, DRRM Expert



Mr. Dhungel attributed the recent fire in Madhav Narayan to the dense settlements, poor planning, and a lack of awareness. To prevent similar incidents, he emphasized the importance of constructing fire-resistant houses and adding fire-resistant layers to roofs. He stressed the need for adequate fire fighting equipment, preparedness, and prepositioning strategies. Empowering local communities to manage and control fires and developing comprehensive settlement plans are also crucial. Mr. Dhungel highlighted the necessity of linking financial resources from initial response to long-term recovery, focusing on integrating Water, Sanitation, and Hygiene (WASH) components. He advocated for incorporating Building Code Implementation (BCI) into development projects and appropriately zoning settlements, including establishing fire code systems to mitigate fire risk. Encouraging research with youth participation and fostering openness from the NDRRMA for data-driven preparedness and multi-hazard response are also essential steps.

Panelist: Mr. Sundar Prasad Sharma, Under Secretary, NDRRMA

Mr. Sharma emphasized the importance of collaboration between DRRM stakeholders and the government to address fire risks effectively. He highlighted the proximity of settlements and community forests, stressing the need for local responsibility in fire prevention and response. The NDRRMA is actively coordinating efforts to enhance fire management and safety through comprehensive training programs for responders and security forces. He announced that a framework is being developed by NDRRMA to support these initiatives, focusing on long-term capacity building and institutional strengthening. NDRRMA has provided fire fighting equipment like high-efficiency portable pumps to municipalities in the Terai region aiming to improve local firefighting capabilities and reduce response times. Mr. Sharma called for integrating fire safety into development plans, advocating for fire-resistant construction and materials as standard practices. Continuous collaboration among DRRM stakeholders is essential to build resilient infrastructure and promote a culture of safety and preparedness across Nepal.



Panelist: Ms. Parbata Gautam, General Secretary, FECOFUN (Federation of Community Forestry Users Nepal)



Ms. Gautam highlighted the severe impact of fires over the past year, which resulted in 124 deaths, 624 injuries, flight disruptions, and eye problems. She emphasized the importance of ongoing awareness efforts while maintaining the reputation of community forests. Minimizing forest and wildfires is crucial, particularly as rising temperatures and increased dryness make small sparks leading to forest fires. Communities are the first responders in fire control, and local bodies are the initial point of contact during fire incidents. Therefore, Ms. Gautam stressed the need for proper firefighting equipment, warehouses, and capacity building for local frontline forest watchmen. Preparedness and prepositioning are essential, and local resources should be optimized while adhering to fire control regulations. Coordination and budget allocation by local governments and forest offices for fire control are necessary, with civil society acting in parallel. She advocated for the creation and implementation of a Community Forest Act and suggested using community ringtones to disseminate fire prevention messages from local radio messaging and jingles. She highlighted for raising public awareness, protecting public ponds, and managing water conservation as the essential preparedness measures at community level.

Concluding remarks by Mr. Machakaji Maharjan, Deputy Director General, DUDBC.MoUD

Mr. Maharjan appreciated the critical discussions on fire risk hazards and preparedness. He outlined proactive response and recovery measures of the DUDBC in addressing these risks through initiatives like the People's Housing Program and Safe Citizens Housing Program. Further, the DUDBC aims to replace vulnerable straw houses with safer alternatives like corrugated galvanized iron sheets or concrete roofs, alongside updating electrical and fire safety codes and introducing a new national building code for fire safety. Policy-level interventions are crucial for widespread adoption of these codes, supported by rigorous implementation at local levels. Mr. Maharjan stressed the comprehensive approach to minimize fire impacts, enhance community safety, and build resilience through fire-resistant infrastructure, integrated WASH components, and flexible policies. Capacity building, community engagement, and coordinated efforts are essential for effective fire management and mitigating forest fires, supported by proper apparatus and policies like the Community Forest Act.



Key takeaways message of the session

- Need of the research and technology-driven solutions for air pollution connected to forest fires
- DUDBC's comprehensive approach aims at risk identification, data-driven strategies, and policy interventions for community safety and resilience.
- Emphasized local awareness and preventive measures against fires, community engagement and urging collaboration among all concerned at community level enhances the firefighting capabilities.
- Policy and building code updates, constructing fire-resistant infrastructure, integrating WASH components, including provisions for audits and mandatory safety measures, are suggested for better fire risk preparedness and management.
- Mitigating forest fires requires proper apparatus and coordination, alongside proper implementation of policies like the Community Forest Act.

Youtube: <https://bit.ly/47etotv>





4.4 Panel Session: Enhancing Multi-Hazard Disaster Risk Reduction Approaches and Technologies for Resilient Shelter

Session Facilitator: Dr. Ramesh Guragain,
Deputy Executive Director, NSET



Session Chair: Mr. Rameshwor Marahatta,
Joint Secretary, NDRRMA

This panel discussion aimed to foster support and collaboration among stakeholders in Nepal's disaster risk reduction sector. The Session Facilitator focused on identifying and prioritizing cost-effective and

feasible technologies tailored to Nepal's context. Government experts from the Department of Hydrology and Meteorology (DHM), the Department of Mines and Geology (DGM), and the Department of Water Resources and Irrigation (DWRI) discussed enhancing multi-hazard disaster preparedness through early warning systems, proper use of meteorological and geological data, and hazard mapping to inform decision-making. The facilitator explored international and national technologies for mitigating landslide risks, integrating water-sensitive designs in flood-prone areas, and ensuring safer shelter construction. Additionally, discussions included the World Bank's innovative financing mechanisms and partnerships to strengthen shelter resilience in vulnerable communities.

Panelist: Dr. Archana Shrestha, Deputy
Director General, DHM

Dr. Shrestha outlined the extensive role of DHM in providing crucial health, energy, climate, and monsoon-related data along with a pivotal role in issuing early warnings and disaster alerts through SMS in collaboration with NDRRMA. DHM also supports climate research by providing essential



data, aiding researchers in tackling climate-related challenges. DHM is actively engaged in capacity-building efforts and works closely with stakeholders



to enhance data utilization. The focus includes improving observation, forecasting capabilities, and conducting relevant research aligned with local governance frameworks by collaborative data sharing, utilizing ICT tools for data tracking and public awareness, strengthening early warning systems, and jointly enhancing disaster mitigation and institutional capabilities.

Panelist: Ms. Monika Jha, Deputy Director General, DMG

Ms. Jha highlighted DMG's focus on mitigating two primary hazards, earthquakes and landslides. DMG places a strong emphasis on using geological data and hazard mapping to support informed decision-making processes for settlement planning, city planning, and infrastructure development. The department actively monitors landslides, conducts earthquake research, and engages in real-time monitoring through data coding and precaution studies. She emphasized that potential users of DMG's information include infrastructure

developers, research institutes, academics and researchers who use the data for analysis and map preparation. Since 1986, DMG has been involved in preparing data and maps through zoning and conducting landslide studies. Over the past five years, the department has dedicated efforts to inventory mapping and providing susceptibility maps, with a particular focus on landslide mapping. However, DMG faces significant challenges, particularly in securing financing, which impacts its ability to concentrate on regions like Karnali and Sudurpashchim provinces.

Panelist: Mr. Prem Lasiwa, Senior Divisional Engineer, DoWRI/Ministry of Energy, Water Resources and Irrigation (MoEWRI)

Mr. Lasiwa detailed the efforts of DoWRI/MoEWRI in managing water-related hazards like floods and flash floods. Operating across multiple offices and managing major rivers, DoWRI/MoEWRI focuses on flood control, erosion mitigation, and river training. Further, collaboration is also done with stakeholders to develop guidelines for watershed and river basin management, particularly in the Terai region, using long-term strategies for flood management. Highlighting sustainable water



supply solutions, Mr. Lasiwa emphasized initiatives like rainwater harvesting and decentralized water treatment systems integrated into shelter planning. These efforts aim to provide disaster-affected communities with reliable water sources. By incorporating water-sensitive design principles and advanced technologies, DoWRI/MoEWRI seeks to bolster community resilience against water-related hazards across Nepal's varied terrain and climate. Their approach includes stakeholder collaboration, technological innovation, and strategic long-term planning.

Panelist: Mr. Basanta Paudel, Disaster Risk Management Specialist, World Bank Nepal



Mr. Paudel outlined the World Bank's initiatives aimed at enhancing shelter resilience in vulnerable communities across Nepal, emphasizing their commitment to integrating global best practices in country-specific disaster risk reduction. Drawing on international experiences, the World Bank focuses on adapting advanced technologies and methodologies to build shelters capable of withstanding earthquakes, floods, and other hazards and also promotes the global-local exchange of knowledge, support green resilient initiatives, and advocate for policy transformations, including contributions to Nepal's National Building Code. Highlighting the role as a knowledge bank, Mr. Paudel emphasized the efforts of World Bank to transfer expertise from global contexts to strengthen Nepal's disaster risk management. He cited the successful application of lessons learned

from the 2015 earthquake in Nepal to manage floods in Pakistan as an example. Collaborating closely with DUDBC and NDRRMA, the World Bank provides technical support to promote resilient construction practices and innovative financing mechanisms for disaster risk reduction in Nepal.

Concluding remarks by the Session Chair, Mr. Rameshwor Marahatta, Joint Secretary, NDRRMA

Mr. Marahatta commended the insightful contributions from panelists, emphasizing the critical need for enhanced coordination and collaboration to foster resilient shelters and reduce disaster risk in Nepal. He underscored the importance of utilizing local materials and adopting GRID principles in construction practices. Moreover, Mr. Marahatta stressed the implementation of the National Building Code to ensure shelters can withstand various hazards effectively. The panel discussion highlighted the urgency of scaling up early warning systems and disaster alerts while facilitating knowledge exchange on resilient shelter solutions. Discussions focused on identifying cost-effective and feasible technologies suitable for Nepal's diverse contexts. The overarching goal is to empower communities in disaster risk reduction efforts and establish sustainable partnerships for resilience-building at the grassroots level.



Takeaways message of the session:

- The DHM plays a critical role in providing data on health, energy, and monsoons, sharing early warnings, and supporting climate studies. It focuses on improving observation, monitoring, forecasting, and research capabilities.
 - The flood management initiatives undertaken by DoWRI focuses on flood and erosion control, river management, and sustainable water supply solutions like rainwater harvesting and groundwater recharge to enhance shelter resilience.
 - The World Bank emphasizes adapting international experiences to Nepal's context, promoting resilient construction practices, and supporting disaster risk reduction through innovative financing and partnerships.
 - Need to utilize local materials and adopt GRID in construction underscores to ensure shelters to withstand multiple hazards.
- Knowledge exchange and best practices sharing provides a way forward to foster collaboration among stakeholders for resilient shelter solutions and establishing partnerships for sustainable resilience-building at the grassroots level.

Youtube: <https://bit.ly/4e66cAB>



4.5 Panel Session: Integrated Approaches to Reconstruction and Recovery

**Session Facilitator: Ms. Radha Verma,
Senior Partnership Officer, UNOPS**



**Session Chair: Mr. Anil Pokhrel,
Chief Executive, NDRRMA**



The panel session focused on discussing a multifaceted strategy for national rebuilding following the 2015 Nepal earthquakes, emphasizing the need for a comprehensive and coordinated effort among government agencies, international organizations, and local communities. The Session Facilitator Ms. Radha provoked the panelists to share best practices in sectors such as infrastructure, housing, livelihoods, healthcare, and education. The discussion highlighted the importance of both immediate recovery and long-term resilience. Challenges in localizing integrated approaches were explored by the Facilitator, alongside international and regional strategies for post-disaster recovery. The session also examined the expanding role of the private sector in promoting sustainable and locally viable shelter construction solutions.

**Panelist: Dr. Chandra Bahadur Shrestha,
Former Executive Committee Member,
NRA**

Dr. Shrestha mainly highlighted the two primary policy frameworks used in Nepal's post-2015 earthquake recovery ie. the PDNA and the Act on Reconstruction of Earthquake Affected Structures, 2015 that supported the former NRA to develop approximately 23 to 24 manuals and working procedures. Dr. Shrestha highlighted the challenges in recovery due to the prioritization of private housing reconstruction and insufficient flexibility within government mechanisms. He pointed out the need for longer gestation periods for recovery efforts and better integration of livelihood opportunities into reconstruction plans for current disaster recovery programs. Dr. Shrestha stressed that the context in Nepal has evolved, with disaster mitigation and resilience responsibilities now primarily resting with local governments under the federal system. He recommended restructuring the NDRRMA for more effective decision-making. Dr. Shrestha also called for proper resources allocation to local bodies for better planning and disaster response, better mentoring of local engineers and highlighted the need for fundamental changes to ensure local governments can meet disaster management by shared collaborative responsibility.



Panelist: Mr. Sudhir Kumar, Risk Management Specialist, UNDP Nepal



Mr. Sudhir underscored the importance of distinguishing between relief, recovery, and reconstruction, noting that while relief efforts are short-term and politically charged, recovery and reconstruction require long-term commitment and flexibility. He also defined an integrated approach through three key terms: temporal integration, sectoral integration, and policy/administrative integration citing suitable examples of Rohingya refugee shelters in Bangladesh, Gujarat earthquake recovery and other international examples. For practical recommendations, Mr. Sudhir emphasized the need for clear policies, strong leadership, and partnerships with development organizations. Clear policies help prevent unplanned reconstruction and secure commitments from development partners. Strong leadership is vital, with historical examples highlighting key figures who drove successful recoveries. Engaging development partners is crucial, as governments alone cannot meet all needs, especially in specific cases like orphaned children during the Gujarat earthquake recovery. Partnerships with community-based organizations can address such gaps effectively. In summary, clear policies, strong leadership, and robust partnerships are essential for successful integrated recovery and reconstruction.

Panelist: Mr. Björn Söderberg, Founder and Director, Build up Nepal

Mr. Söderberg emphasized the need for proper technologies that local people can adopt and strict adherence to building codes. Söderberg stressed the importance of locally led reconstruction, advocating for the use of local materials and the establishment of small enterprises to produce these materials. Training local masons and engineers is crucial to ensure compliance and build a resilient ecosystem. He pointed out the environmental impact of traditional reconstruction methods and praised the new policy favoring green construction materials, urging for its thorough implementation. For greater involvement of the private sector and local stakeholders, Söderberg recommended engaging them early in the project design phase to leverage their efficiency and expertise. He called for the government and development organizations to lead by example by using green materials in their buildings. Reflecting on lessons from the 2015 earthquake, he emphasized the need to prepare the supply chain, build local capacity to prevent price surges and ensure quality construction, along with training to local masons and capacity exposure to engineers to enable compliant and environmentally friendly rebuilding, making the policies effective and practical.



Panelist: Mr. Gaurav Panthi, Program officer, JICA Nepal



Drawing from JICA's experience post-2015 earthquake, Mr. Panthi discussed the comprehensive approach that combined reconstruction with DRR actions like risk mapping and capacity development for government agencies and local communities. He stressed the significance of community participation and the involvement of trained local masons, which accelerated the reconstruction process and created local jobs. JICA's approach also involved integrating various support schemes, such as concessional loans, grants, and technical cooperation, to cover different reconstruction and DRR activities, ensuring a holistic and sustainable recovery effort. Looking forward, JICA is shifting its focus from recovery-reconstruction to disaster risk reduction, aligning with the Sendai Framework and the concept of building back better. Mr. Panthi outlined JICA's revised cooperation plan for Nepal, which prioritizes DRR and climate change, emphasizing pre-disaster investment as a key strategy to mitigate future economic losses. Strengthening DRR governance and promoting pre-disaster investments are central to JICA's current initiatives as the ongoing projects with various Nepalese government agencies that focus on ensuring DRR measures are integrated into municipal planning and development strategies.

Concluding Remarks by the Session Chair Mr. Anil Pokhrel, Chief Executive, NDRRMA

Mr. Pokhrel shared delights over praiseworthy discussion on integrated approaches. He shed light on the significant progress made over the last eight years in building over 800,000 houses and thousands of schools post the 2015 earthquake. He equally stressed for lessons learned from reconstructing houses damaged by smaller-scale disasters to larger ones in Nepal and also crucial to integrate these experiences into a comprehensive recovery strategy. He advocated for a holistic approach that encompasses multiple hazards, not duplicating new Standard Operating Procedures for each hazard but to streamline the reconstruction process, with a strong emphasis on the involvement of municipal governments and the use of local and green focused nature-based materials to enhance resilience and generate local employment. He highlighted the need for a single and mission-oriented integrated reconstruction strategy, similar to the 2015 Nepal earthquake approach to address West Nepal earthquake. He underscored the importance of timely financing, preserving indigenous construction methods, and ensuring the capacity gains by municipalities to lead reconstruction efforts. Pokharel also stressed the importance of comprehensive risk assessments, universal access designs, collaborative approach and the need for effective communication strategies to sensitize and engage communities.



Key takeaways message of the session:

- Under Nepal's federal system, disaster mitigation and resilience responsibilities have shifted to local governments, which face significant challenges due to limited financial and human resources.
- Recommendations for restructuring the NDRRMA for better decision-making and stressed the need for fundamental changes to support local governments.
- Advocacy for an integrated approach involves temporal, sectoral, and policy/administrative integration, citing examples like the Rohingya refugee shelters and Gujarat earthquake recovery.
- Locally led reconstruction compounded with usage of green materials along with training local masons and engineers promotes resilience and sustainability.

- Recovery should also reduce vulnerability and enhance resilience, thus combining reconstruction with DRR actions, community participation, and the integration of support schemes.

Youtube: <https://bit.ly/3MxhPUH>



4.6 Panel Session: Leveraging ICT Engagement for Disaster Preparedness, Response and Recovery

**Session Facilitator: Mr. Sachin Malego,
National Information Management
Officer, NHSRP**



**Session Chair: Mr. Krishna Hari Banskota,
Former Chief Information Commissioner
of Government of Nepal**



This panel session aimed to deepen understanding of ICT's role in disaster management, focusing on shelter provision from preparedness to long-term recovery. Session Facilitator Mr. Sachin, highlighted best practices and lessons from academic research, policy initiatives, and practical experiences with technologies like Artificial Intelligence (AI), drones, and blockchain. The session emphasized collaboration and knowledge sharing among diverse stakeholders to promote ICT-driven solutions for more effective and equitable disaster

response and recovery. Additionally, the facilitator explored the roles of government and non-government agencies in using ICT to enhance community participation, digital literacy, and transparency through management information systems and public information sharing. Further, contributions of academic research into actionable policy recommendations was also inquired by the facilitator that highlighted the regional initiatives such as Regional Integrated Multi-Hazard Early Warning System (RIMES) in developing data integration systems and data sharing mechanisms for enhanced data access and insights for early warning response.

**Panelist: Dr. Iguniwari T. Ekeu-wei,
Disaster Risk Management Specialist at
SAR Climate Change and DRM (SSACD),
World Bank.**

Dr. Iguniwari emphasized the crucial role of data and technology in disaster management, and noted that Nepal faces challenges with data uniformity due to varied collection methods by different entities, complicating integrated assessments and data sharing. He praised the NDRRMA's Bipad portal for data visualization but stressed the need for further development to enhance analytics and decision-making. Standardizing data collection and improving national geospatial information systems are key steps forward. He discussed the use of satellite and drone technology for damage assessment and the benefits of anonymized cell phone data for tracking population movements post-disaster. Dr. Iguniwari also highlighted the importance of leveraging initiatives like RIMES and the International Disaster Charter for better



collaboration and resource access. He called for policymakers to reduce bureaucratic barriers and build partnerships to harness ICT's potential in disaster preparedness and response, with support from international bodies like the World Bank. In shelter planning and management, he advocated for ICT tools to enhance community engagement, resilience, sustainability, public-private partnerships, and environmental risk mitigation. He also cautioned about potential challenges and unintended consequences of emerging technologies like virtual and augmented reality.

Panelist: Dr. Uttam Paudel, Program Director- DRR and Climate Change Adaptation, CRS Nepal

Dr. Paudel emphasized the ethical and effective use of ICT-driven approaches to enhance community participation, timely needs assessment, and



adequate housing solutions for disaster-affected people. He highlighted the importance of involving both policymakers and communities to bridge the digital divide, improve communication strategies, and make informed policy decisions. Despite 98% of Nepalis using mobile phones, only 31% engage in specific digital activities like mobile banking, indicating progress but also a significant gap. Challenges include communication barriers and data complexity, which hinder access to disaster-related information. Dr. Paudel pointed out the issues faced during the 2015 post-earthquake recovery, such as fraudulent reports of affected households and data duplication that emphasized the need for accurate data representation in a systematic manner. He called for scaling up communication efforts to

make disaster information more accessible and understandable. By leveraging lessons from past experiences, Nepal can enhance its disaster management strategies, ensuring resilience and effective implementation.

Panelist: Mr. Buddhi Sagar Shrestha, Director, Disaster Management Department, IFRC

Mr. Shrestha highlighted the evolution of the NRCS from initial relief efforts in the 1960s and 1970s to its current focus on Community Based Disaster Risk Reduction (CBDRR) and anticipatory actions.



He emphasized the integration of digitization in disaster management, enabling proactive measures like weather forecasting and early warning systems. Key initiatives included deploying volunteers for data collection using the Bipad portal, conducting detailed damage assessments with the Kobo survey toolkit, and ensuring timely, efficient, and data-driven responses. NRCS's notable efforts during the Jajarkot Earthquake response included initial rapid assessment conducted by around 2,000 volunteers. Dr. Shrestha stressed the importance of enhancing registration and tracking systems, addressing potential inequalities, and improving data management and ownership. Despite initial ownership issues, other organizations have started using the data collected by NRCS. Mr. Shrestha underscored the need for suitable infrastructure development in remote areas to support ICT integration, ensuring it reaches vulnerable and marginalized communities. He also highlighted the importance of detailed surveys, images of affected areas, and data-driven decision-making to address community needs effectively.

Panelist: Ms. Puja Shakya, Nepal Country Program Lead, Regional Integrated Multi-Hazard Early Warning System (RIMES)



Ms. Shakya outlined the focus areas of RIMES on providing technical services on hydrometeorology services across regions like Nepal, Southeast Asia, South Asia, and Africa as well. RIMES addresses the data gaps and sharing issues through platforms like DataX, which facilitates the storage and sharing of climate-related data regionally. Nepal contributes the Hydromet data to DataX, supporting the development of weather and climate forecasting tools. Ms. Shakya underscored a challenge of insufficient data for accurate forecasts, which RIMES addresses through regional data sharing on platforms like DataX. Operational challenges persist in connecting national-level systems to local levels for effective information dissemination. Meanwhile during the conversation, Ms. Shakya identified a co-development approach with like-minded government agencies in capacity building collaboration initiatives to forward decision support systems by involving key IT focal persons. She informed participants on strengthening capacities of DHM through training, secondments, and model development, exemplified by developing a flood model for the Babai River in collaboration with DHM as the leading example of collaborative approach. She added that RIMES is committed to supporting both regional and local-level activities whereby readiness to work on tailoring methods as required.

Concluding remarks on Session Chair, Mr. Krishna Hari Banskota, Former Chief Information Commissioner of Government of Nepal

Mr. Banskota applauded the insightful contributions to the panel discussion on leveraging ICT for disaster management and recovery thus recognized diverse expertise and valuable perspectives of the panelists. He emphasized on data uniformity, collaborative approaches, capacity building, and the pivotal role of technology in disaster resilience and response among all like-minded government and non-governmental stakeholders. Mr. Banskota further nailed the collaboration between regional organizations, government agencies, and stakeholders and underscored its significance in addressing challenges and maximizing the potential of technology for effective disaster management. He encouraged call to action on showcasing continued efforts in digitalization, investment in ICT infrastructure, and policy changes to enhance disaster resilience and response, urging to translate ideas into actionable strategies. Mr. Banskota emphasized inclusivity, sustainability, and the need for concerted efforts in leveraging ICT for disaster management, while acknowledging the challenges involved in implementing effective solutions.



Key takeaways message of the session:

- Importance of addressing infrastructure vulnerabilities in promoting ICT should be included in training programs for technical personnels.
- Proper utilization of data and ICT technologies boosts the collective efforts in building resilient communities and addressing the complexities of disaster management.
- Capacity building involves knowledge transfer and collaboration during operations, learning from best practices and insights from other regional disaster management efforts to inform future plans.
- Regional collaboration with government agencies and other organizations working in Nepal to collectively address ICT-related challenges in disaster risk management.
- Sharing forecast information developed at the

national level with regional-level stakeholders for analysis and dissemination to end users, conducting climate application platforms and forums to facilitate understanding and preparedness.

Youtube: <https://bit.ly/47c5UW5>



4.7 Panel: Fostering Inclusive Resilience by integrating GEDSI

Session Facilitator: Mr. Gajendra Budathoki, Journalist



Session Chair: Ms. Abha Shrestha Karna, Joint Secretary, MoWCSC

This panel session aimed to foster thoughtful discussions on integrating the GEDSI component into DRRM to ensure communities can effectively prepare for, respond to, and recover from disasters



and other shocks. Session Facilitator, Mr. Gajendra, engaged the panelists in identifying the unique challenges faced by vulnerable groups, including women, people with disabilities, and marginalized communities, in the context of disasters. He purposefully guided the discussions to envision the integration of GEDSI perspectives into planning and implementation processes and extensively sought recommendations to promote inclusive resilience.

The session's thought-provoking environment and participant feedback highlighted the importance of including the voices of vulnerable and marginalized people. This provided opportunities for knowledge exchange, collaboration, policy influence, and community engagement, enabling stakeholders to work together to build more resilient, equitable, and inclusive communities that are better prepared to withstand and recover from disasters and other shocks.

Panelist: Ms. Reena Chaudhary, GEDSI Focal Point, NDRRMA

Ms. Chaudhary elaborated the significant challenges faced by different vulnerable groups that are not adequately addressed in the country's disaster management discourse. She highlighted the necessity for the significant inclusionary policy systems to effectively address the needs of vulnerable groups of people. She further added on the cruciality of integrating GEDSI strategies into planning and implementation processes. Notably, the NDRRMA has developed a Disaster Risk Reduction and Management Act 2017, Disaster Risk Reduction Strategic Plan of Action 2018-2030, and relevant other procedures and guidelines that align with these inclusive principles. Recently, the NDRRMA endorsed the GEDSI Strategic Action Plan 2081 on disaster management, which was formulated with reference to the Sendai Framework, the Build Back Better approach, and the Sustainable Development



Goals (SDGs). Ms. Chaudhary expressed optimism that these measures will address various social issues pertaining to disaster risk reduction, inclusion of vulnerable groups in disaster response and recovery and beyond. She noted that Nepal's constitution and laws emphasize GEDSI, ensuring that policies are gender-friendly and inclusive that truly underscores the importance of these measures in fostering a more inclusive and resilient approach to disaster management in Nepal.

Panelist: Ms. Jeevika Khadka, Program Manager-DRRM, Youth Innovation Lab

Ms. Khadka discussed the significant issue of data availability and the existing gaps (gender and vulnerability data) that hinders effective disaster management and inclusiveness. She stressed that



data plays a crucial role in managing any type of disaster from GEDSI perspective at the local level. There is a notable lack of initiatives for data collection, which calls for critical need for strategies, policies, and partnerships to foster inclusive recovery and reconstruction efforts in Nepal. Her remarks directed towards the importance of integrating GEDSI considerations into formulating the resilience policies, plans, programs and activities at local, national, and global levels. Ms. Khadka shared that technology development, open data use, data driven advocacy, shared collaboration, youth engagement, volunteerism and entrepreneurship focused programs are few examples of effective approaches that have successfully mainstreamed GEDSI principles. To address this, she suggested incorporating GEDSI

indicators into data life cycle from collection of household surveys data conducted at local levels and its analysis. This would enable the inclusive collection of disaggregated data, which is essential for reflecting community vulnerabilities, conducting vulnerability assessments, and community mapping for disaster management and for making informed decisions.

Panelist: Ms. Sama Shrestha, Programme Specialist, UN Women

Ms. Shrestha underscored the critical role of both local and international partnerships in addressing GEDSI issues within disaster management efforts in Nepal. She applauded the recent endorsement of the GEDSI Strategy Action Plan 2081 by the Nepal Government and stressed on the importance of its implementation at local level effectively. For successful implementation of the GEDSI Strategic Action Plan, Ms. Shrestha pointed out the necessity of community-centered and people-centered partnership approaches. She emphasized that local involvement in planning is crucial, and international and private organizations should support the government in these efforts. Utilizing methodologies that focus on community and individual needs ensures participation at all levels and helps identify vulnerable communities, people with disabilities, and other at-risk groups. Ms. Shrestha also pointed out that GEDSI partnership programs need to advance for greater effectiveness. For relief programs, she advocated for the mobilization of community volunteers as one of the best implementation strategies for disaster management, ensuring grassroots involvement and tailored support for affected communities.



Panelist: Mr. Sugam Bhattarai, General Secretary, National Federation of Disabled Nepal (NFDN)



Mr. Bhattarai applauded the significant progress made in integrating disability considerations into disaster management programs through NDRRMA's interventions that has listed accessible programs for individuals with disabilities, marking a positive step towards inclusive program implementation by the government. Mr. Bhattarai shared a successful example of advocacy by NFDN, which influenced housing design discussions after the 2015 Nepal earthquake. Their efforts resulted in the incorporation of ramps and larger doors in school buildings, demonstrating a noteworthy advancement in inclusive building construction. As the national cross-disability umbrella organization, NFDN has consistently coordinated and advocated with NDRRMA during the formulation of policies and laws, providing feedback and suggestions to ensure disability inclusion in construction and disaster management. NFDN's involvement extends to all levels of policy dialogue and policymaking with NDRRMA, focusing on inclusiveness in disaster management and continuing partnership and advocacy efforts for underscoring the importance of disability inclusion, especially in the face of climate-induced disasters.

Concluding Remarks Session Chair, Ms. Abha Shrestha Karna, Joint Secretary, MoWCSC

Ms. Karna shared brief highlights of the vibrant discussions around integrating GEDSI in DRRM while concluding the panel session. She emphasized the need to foster inclusive resilience by integrating GEDSI into the socio-political system. She highlighted the importance of inclusive GEDSI data and pointed out existing data gaps. Addressing these social issues requires an intersectional approach, particularly in terms of accessibility. Ms. Karna stressed the need for designing activities that involve people with different abilities, making them responsible and accountable while ensuring they benefit from their participation. She acknowledged the government's commitments to national and international conventions, noting that notable progress has been made towards these commitments, particularly with the GEDSI Strategic Action Plan 2081. She called for the active participation of all stakeholders in implementing programs for GEDSI integration.



Key takeaways message of the session:

- Significant advancements and breakthroughs are anticipated through the implementation of the NDRRMA's GEDSI Strategic Action Plan 2080 in forthcoming days.
- All stakeholders must commit to making significant investments and implementing GEDSI strategies in relevant ways. This will ensure that the needs of vulnerable groups are met, and they are treated equally during times of disaster.
- The absence of proper data management at the local level results in difficulties in identifying vulnerable communities and groups which hinders effective response to them.
- Emphasizes the need for pertinent and precise data to be obtained, to the best of our ability, in order to ensure efficient disaster management.
- For the relief program, mobilizing community volunteers is one of the best implementations approaches for disaster management.

Youtube: <https://bit.ly/3zeJZAQ>





**Panelist: Mr. Bijay Prakash Wanem,
Chairperson Maiwakhola Rural
Municipality, Taplejung district**



Chairperson Wanem addressed the severe loss and damage caused by floods and landslides in June 2023 in Maiwakhola Rural Municipality, Taplejung district. He drew the attention of participants on the increasing risk of multiple disasters and harsh climatic conditions in Nepal. As an example, he mentioned the irregular bloom of 28 species of rhododendron flowers in Taplejung areas that used to blossom annually, thus anticipating the ecological disruptions. He attributed unpredictable rainfall patterns as the climate induced disaster linking with haphazard settlements established without proper geohazard and environmental impact studies. He also took references of flash floods in Manang and Mustang some years back and also the similar kinds of flood in Kathmandu city as further evidence of the detrimental effects of climate change. In response to the crisis, Mr. Wanem elaborated with development organizations on the municipality's efforts to relocate temporary shelters for the 65 families who lost their homes. He emphasized the importance of planned settlement and environmental considerations in mitigating the impact of future disasters.

**Panelist: Mr. Narayan Prasad Bhandari,
Deputy General, DUDBC/MoUD**

Mr. Bhandari informed the participants about MoUD's efforts in currently revising its policies related to enhancing the resilience of urban settlements and collecting feedback from discussions like this one. He highlighted the ongoing efforts to update the national building code to ensure climate-resilient building designs and infrastructure, while he touched upon Janata Awas Program and Surakshit Nagarik Awas Program that addresses the housing needs of people affected by disasters. Mr. Bhandari stressed the importance of maintaining green solutions and alternative energy application for a balanced ecosystem, and risk-sensitive land use planning to minimize the potential risks of climate-induced disasters. Mr. Bhandari called for future strategies and needful collaborations that secure settlements and infrastructure against these challenges. He pointed out the Ministry's proactive steps in responding to climate change impacts through formulation of National Urban Development Strategy 2017 and promoting the programs such as Integrated Urban Development and Urban Infrastructure Development in selected municipalities.



Panelist: Dr. Maheshwor Dhakal, Joint Secretary, MoFE



Dr. Dhakal addressed the significant impacts of climate change on housing and settlements by highlighting the melting of glaciers and potential outburst floods. He outlined examples of flood inundation in Terai region leading to substantial property damage annually. Other examples were, drying of aquifers and water wells in some villages, forcing residents to travel long distances for water, resulting in both economic and intangible losses. He underlined the irreplaceable and immeasurable nature of human losses due to disasters. He mentioned the establishment of the Loss and Damage Fund at Conference of Parties -COP27, which was operationalized at COP28. He serves as an Alternate Board Member for this fund as representative from Nepal Government which initially started with USD 601 million. The first board meeting in the United Arab Emirates discussed mainly on fund management and fund disbursement strategies. He stressed the need for accurate measurement of losses and damages to claim appropriate compensation, highlighting the technical challenges in economic analysis. Dr. Dhakal suggested the need of mobilizing youths for these measurements thus promoting employment and referred to comparative studies indicating Nepal's losses are around 3.89% of GDP. He concluded by calling for the integration of these considerations significantly into government programs and activities.

Panelist: Dr. Bimala Rai Poudyal, Former Minister of Foreign Affairs (MoFA)

Dr. Rai Poudyal addressed the socio-structural dimensions of climate-induced loss and damage, of the impact at individuals and family level due to resettlement and relocation, emphasizing the importance of considering intersectionality, differentiated risk, and vulnerability in responding to climate change impacts. She also highlighted the differential impacts on housing and settlements, noting that these spaces encompass cultural, traditional values, and social norms central to people's lives. She revealed gaps of disaster preparedness in PDNA of 2015 Nepal and advocated for effective settlement management that should prioritize reducing risks for vulnerable groups such as senior citizens and pregnant women who may share living spaces during emergencies. Dr. Rai Poudyal mainly emphasized the importance of inclusive preparedness, leveraging community innovation and local practices to combat climate impacts. She advocated for building codes that address inclusiveness and the health implications for women, children, and senior citizens. Regarding the role of parliamentarians and legislators, Dr. Rai Poudyal called for their active involvement in influencing debates and discourses addressing the scale of loss and damage. She recommended that one-third of participants in any capacity-building training should be female to ensure gender inclusiveness and equitable representation.



**Concluding remarks by Session Chair,
Dr. Arnico Pande, Member of National
Planning Commission (NPC)**



Dr. Pande praised the worthy discussions around climate induced loss and damage. He applauded the crucial role of the government in fostering climate-friendly, economically efficient, and timely construction of infrastructures through the Nepal reconstruction program. Dr. Pande pointed out that climate-induced disasters pose an escalating and cascading risk of severe impacts on human lives that calls for comprehensive strategies to minimize these risks. In illustrating the current climate change scenarios, Dr. Pande provided context by discussing the changing temperature patterns by citing examples of Dengue as a common occurrence in Kathmandu, signifying a broader shift in public health challenges. Additionally, he mentioned the upward shifting of the tree lines, which has cascading effects on balancing ecosystems and minimizes the climatic heat patterns in local communities. Dr. Pande's remarks underscore the urgent need for proactive government intervention at all levels effectively and the implementation of sustainable practices in the housing and settlement sector to combat the adverse effects of climate change.

Key takeaway messages of the session:

- Mapping of national and international funding mechanisms to be tracked to efficiently mobilize resources for mitigating loss and damage.
- Prioritize disaster preparedness along with identification and implementation of disaster risk reduction interventions.
- Emphasize the anticipatory actions and well-equipped stockpiling of resources to enhance effectiveness in post-disaster response.
- Implement early warning systems, forecast-based risk reduction and management programs from national to local level.
- Enhance the capacity of all responders and volunteers involved at the local level to assure well trained readiness in disaster response while fostering inclusiveness.

Youtube: <https://bit.ly/4cWvC27>





4.9 Panel Session: Exploring Innovative GRID Practices for Resilient Shelter, Housing, and Settlements

Session facilitator: Ms. Preeti Karki, GRID and Partnership Technical Coordinator, NHSRP



Session chair: Dr. Sangeeta Singh, Member of National Planning Commission (NPC)



This panel session was organized to evaluate the progress made by the GRID initiatives since its endorsement in 2021 by Nepal government and to examine ongoing developments following

the GRID pathways. The discussion purposefully aimed to bring together various stakeholders in relation to taking forward green, resilient and innovative solutions in shelter and settlements and infrastructures development. The Session Facilitator Ms. Preeti probed the panelists on unpacking the commitments of concerned Ministry level and directions of development partners in planning the GRID initiatives from shelter to settlements and also in sustainable land use and relocation aspects. Facilitator also inquired on the exemplary best practices of using the green, locally available and affordable technologies by taking feedback from Habitat from Humanity that has revised its country strategy to incorporate GRID principles. Overall through discussions, the Facilitator explored the potential impacts of Green Building Guidelines, GRID Strategic Action Plan (GRID SAP) and other similar policies in addressing green friendly housing technologies, affordability, and inclusive design while aligning with climate change and adaptation.

Panelist: Mr. Stephen Steve Danyo, Program Leader- Sustainable Development, World Bank Nepal

Mr. Danyo highlighted the progress of the GRID



initiatives since the Kathmandu Declaration on GRID was jointly endorsed by the Government of Nepal and 16 development partners, including the World Bank, in September 2021. The GRID initiative represents a \$10 billion partnership focused on ten priority transitions, such as renewable energy, livable cities, and clean air, aiming to boost sustainable development through increased investment. Mr. Danyo informed that the GRID SAP led by the Ministry of Finance is in its final stages, and that it will provide a comprehensive roadmap to achieve GRID objectives, emphasizing urban livability, air quality improvement, and sustainable transportation. These efforts are expected to significantly impact Nepal's housing sector by integrating sustainability and resilience into urban planning and infrastructure development. He also outlined the World Bank's future priorities to support Nepal in achieving GRID targets, extending support to the municipal level. The focus will be on building capacity, providing technical assistance, and equipping municipalities to implement and sustain GRID principles in their development projects, aiming to foster resilience and inclusivity throughout Nepal's housing and settlement sector.

Panelist: Mr. Manoj Nakarmi, Senior Divisional Engineer, DUDBC/MoUD



Mr. Nakarmi provided insights into the interventions of MoUD/DUDBC in housings for disaster affected people and other several urban development programs that focuses on commitment to safety, affordability, and sustainable practices in housing development. Central to these efforts is the development of Green Building Guidelines, aligned with the GRID SAP that aims to meet the

growing demands for environmentally friendly housing with affordable and inclusive designs. Mr. Nakarmi elaborated the characteristics of Green Building Guidelines that covers climate change adaptation, site planning, building design, materials selection, water conservation, waste management, energy efficiency, sustainable practices, and retrofitting of old buildings. Through these guidelines, the MoUD aims to promote sustainable development and contribute to Nepal's vision for green, resilient, inclusive shelter and housing. Mr. Nakarmi underscored the Ministry's commitment to implementing these guidelines henceforth emphasizing alignment with national priorities and calls for stakeholder collaboration in implementation.

Panelist: Mr. Ganesh Prasad Bhatta, Joint Secretary, Ministry of Land Management Cooperatives and Poverty Alleviation (MoLMCPA)



Acknowledging the often overlooked but critical nexus between land tenure security and climate change, Mr. Bhatta highlighted the transition of the GRID initiatives in facilitating relocation to safer land for settlements and enhancing overall land governance to ensure safer shelter options. Mr. Bhatta noted that a study on land administration revealed that 25% of settlements consist of either landless individuals or informal settlers, indicating the pressing need for formalization and management of these areas. To address this issue, the National Land Commission has been established to cater

to the needs of landless and informal settlers, aiming to regulate informal economic activities and integrate them into the formal economy. Furthermore, Mr. Bhatta highlighted the Ministry's efforts to enhance women's access to land through measures such as joint land ownership, resulting in significant progress, with 49.7% of women now owning land according to MoLMCPA data. He noted that collaboration with the World Bank to develop land use plans is underway; however, challenges such as capacity building, financial constraints, and limited technical assistance from development partners persist. Despite these challenges, the Ministry remains committed to aligning with climate and GRID priorities, recognizing the crucial role of effective land management in achieving sustainable development and ensuring safe and secure shelter for all.

Panelist: Ms. Eliza Sthapit, Country Director, Habitat for Humanity Nepal



Ms. Sthapit provided insights into how Habitat for Humanity Nepal has integrated GRID principles into its revised country strategy, emphasizing inclusivity, empowering homeowners through joint land titles in women names too, community resilience, affordability, and the adoption of greener technologies for land and housing construction. She underlined the importance of holistic approaches that intertwine green development with social equity and resilience-building. Moreover, Ms. Sthapit highlighted the dedication of HfH's to construct homes focusing marginalized communities that are resilient to climate-induced disasters, utilizing sustainable materials like treated

bamboo and CSEB. The organization collaborates with the private sector to enhance construction technologies and ensure thermal comfort in homes, aligning with GRID objectives. Additionally, she also highlighted implementation of community-centric approaches such as PASA (Participatory Approach for Safe Shelter Awareness) and Pathway to Permanence in housing projects that prioritizes community involvement and foster long-term stability & sustainable shelter solutions, along with strengthening the capacities of local governments and communities.

Concluding remarks by: Dr. Sangeeta Singh, Member of National Planning Commission, Government of Nepal

Dr. Sangeeta lauded the insightful discussions on GRID initiatives in the face of the need of innovations



and shift towards the nature-based solutions for inclusive development. She highlighted the need to consider sustainability and interventions beyond mere resilience in the face of climate change while following AI and ICT. Traditional cities, including those in Nepal, exemplify sustainable and resilient practices through compact layouts and integrated agricultural and forest land use. Nepal's traditional settlements are governed by unique social norms, such as the presence of the Hindu Goddess Agramatrika, which restricts building expansion. Citing the Melamchi incident, Dr. Sangeeta emphasized learning from traditional practices to avoid mishaps and urged caution in planning large infrastructure projects. She also discussed sustainable transport, housing, waste management, recycling, and reuse, emphasizing the importance

of inclusive social aspects to improve quality of life. Dr. Sangeeta highlighted the 16th Plan of the Nepal Government and added that the settlement planning is approached in an integrated manner, offering numerous opportunities for enhancing the planning process. She concluded by emphasizing the importance of comfort and safety in shelters within the GRID context, particularly adhering to thermal limits to prevent mortality among vulnerable populations.

Key takeaways of the session:

- The GRID Strategic Action Plan (SAP) is in its final stages, focusing on urban livability, air quality, and sustainable transportation and commitment to Kathmandu declaration.
- The Green Building Guidelines, aligning with the GRID SAP aims to reduce energy consumption, greenhouse gas emissions, water usage, and waste production while promoting environmentally friendly, affordable, and inclusive housing designs.
- Establishment of the National Land Commission focuses to address the needs of landless and informal settlers and the significant progress in enhancing women's land ownership.
- Habitat for Humanity Nepal highlighted the organization's integration of GRID principles into its strategy, focusing on marginalized communities and climate-resilient housing through sustainable green materials.
- The 16th Plan of the Nepal Government aims to



Closing Session



The closing session was facilitated by Ms. Ranjana Shrestha, Communication Lead from NDRRMA. The Honorable Home Minister, Mr. Rabi Lamichhane, attended as the Chief Guest, with Special Guests including Mr. Ek Narayan Aryal, Secretary of the Ministry of Home Affairs, Mr. Anil Pokhrel, Chief Executive, NDRRMA and Mr. Azmat Ulla, Head of Delegation from IFRC. Other distinguished guests included Mr. Tapta Bahadur Paudel and Ms. Manisha Singh Tharu, Chairperson and Vice-Chairperson of Rapti Sonari Rural Municipality (Banke district), Mr. Lekh Nath Pokhrel, Advisor to the National Association of Rural Municipalities in Nepal (NARMIN), and Dr. Arnico Pande and Prof. Dr. Sangeeta Singh, Members of NPC. Joint Secretaries of NDRRMA, Mr. Gogan Bahadur Hamal

and Mr. Rameshwar Marahattha, were also present as distinguished guests.

5.1 Summarization of two days event by Mr. Gogan Bahadur Hamal, Joint Secretary, NDRRMA

Mr. Hamal thanked experts, organizers, and participants for their involvement in productive



discussions over the two-day event that mainly highlighted the importance of collaboration and communication among stakeholders to enhance sound efforts of disaster risk reduction, shelter

reconstruction, and recovery efforts. He also praised the worthy discussions in fire risk and preparedness session along with the importance of cross-cutting issues on GEDSI, GRID, and ICT in integrated disaster response and recovery. He noted that the expert's perspectives have enriched the program and stressed the need for disaster preparedness to keep communities' alert. He informed that NDRRMA is developing synchronized DRRM plans from previously fragmented work plans, now addressing multiple hazards beyond just earthquakes. Mr. Hamal urged the enforcement of building codes for safer reconstruction and encouraged local governments to develop their own building codes to design safe houses before disasters occur.

5.2 Appreciation and Token of Love Distribution by Chief Guest



Hon. Deputy Prime Minister and Minister of Home Affairs, Mr. Rabi Lamichhane handed over the Token of Love and the Certificate of Appreciation to representatives of NSF 2024 Organizing Committee members and thanked them for their wonderful contribution in organizing NSF. Similarly, Minister Lamichhane also provided Token of Love to the special and the distinguished guests.

Youtube link of the closing session: <https://bit.ly/3Zgskn1>

5.3 Closing Remarks by the Guests

Closing Remarks by Mr. Azmat Ulla, Head of Delegation, IFRC



Mr. Ulla expressed congratulations to the NDRRMA and the organizers for the wonderful interactions held over the two-day event of the shelter forum as anticipation of opening doors of opportunities in the DRRM sector in Nepal, emphasizing that progress must continue beyond the forum too by taking forward action-oriented solutions and cooperations with like-minded organizations. Mr. Ulla highlighted the complexities of shelter in emergency,



transitional, and long-term contexts and pointed out the solutions lies in solidarity, cooperation, and technical implementation. He underscored the importance of resource management, site suitability, and cultural considerations while developing programs in housing reconstruction, rehabilitation and recovery.. He further added that the IFRC's collaboration with the NRCS and other partners, particularly in response to the Jajarkot earthquake, was noted as a key example of effective action, with cash assistance and empowering communities. Mr. Ulla concluded by stressing the need for inclusive, sustainable, and livelihood-supporting solutions to protect vulnerable populations in Nepal's seismically active regions.

Closing Remarks by Mr. Lekh Nath Pokhrel, Advisor of National Association of Rural Municipalities in Nepal (NARMIN)

Mr. Pokhrel thanked the organizers for the interactive forum and emphasized that disasters like earthquakes strike without warning. He noted that local governments are the first responders in any



disaster, stressing the need to build the capacities of disaster management committees at the Ward and Tole levels for effective management, coordination, preparedness, and response. He highlighted the gap in coordination and communication between central and local governments, urging the central government to address this. Mr. Pokhrel mentioned that NRCS chapters in every district facilitate disaster preparedness and response. He called for differential standards in housing reconstruction policies to accommodate Nepal's diverse geography and advocated for sufficient budget authority for local governments to execute programs effectively.

Closing Remarks by Mr. Ek Narayan Aryal, Secretary of Ministry of Home Affairs

Mr. Aryal highlighted the severe impacts of climate change in Nepal, noting the loss of snow on mountains, increased forest fires, erratic rainfall, droughts, floods, and decreasing water levels. Kathmandu has



faced extreme flash floods, rising temperatures, and water scarcity, while Terai's agricultural production suffers from water shortages. Despite Nepal's financial contributions to international climate efforts, the country faces significant impacts. Aryal called for a comprehensive approach to climate and disaster management, emphasizing the need for better collaboration among Nepal's three levels of government. He stressed the importance of partnerships among government agencies, NGOs, international bodies, and the private sector to pool resources and expertise for successful recovery and reconstruction. Efficient resource mobilization and transparent mechanisms are crucial for building sustainable and resilient infrastructure.

Closing Remarks by Mr. Anil Pokhrel, Chief Executive, NDRRMA

Mr. Pokhrel addressed the closing ceremony of NSF 2024 and highlighted the utmost need of enforcing the building code measures at the community level. With several mechanisms and procedures in place by NDRRMA in recent times, he is hopeful that disaster response efforts are inclusive and accessible to all communities with the increasing level of people's awareness. He further added that we are facing challenges due to the latest budget allocation by

the Ministry of Finance (MoF) which is insufficient for conducting DDA of three earthquakes. For



instance, following the 2015 earthquake, the slow disbursement of reconstruction funds delayed crucial projects. To expedite progress, we propose a streamlined approach where projects are completed within a year, unlike the current 6-7-year timeframe. We've developed a two-year work plan to achieve this goal. Additionally, local communities lack adequate resources and preparedness for disaster response. To address this, we will focus on creating jobs, particularly green jobs, and implementing GEDSI SAP and GRID SAP holistically. We will prioritize the inclusion of disabled individuals and collaborate with the private sector while utilizing local materials in our projects.

Closing Remarks by Mr. Rabi Lamichhane, Deputy Prime Minister/Home Minister

During the closing ceremony, Minister Rabi Lamichhane expressed the exemplary initiatives of NSF organizers and the active enthusiasm of the participants to stay throughout the two-day event. He announced the approval of the amended "Disaster Affected Private Housing Retrofitting, Reconstruction, and Resettlement Tranche Disbursement Procedure 2081", from 22nd Executive Committee Meeting of NDRRMA. This amendment aims to provide housing grants



to those who have lost their homes due to various disasters. Minister Lamichhane reaffirmed the government's commitment to ensuring the right to housing as guaranteed by the Constitution of Nepal. He emphasized the importance of utilizing local resources and technology for future reconstruction efforts, which would create job opportunities and stimulate local economic development. The gathering of government stakeholders and development actors demonstrated their collective commitment to promoting post-disaster recovery, reconstruction, and resilience in the region.

Youtube link of the closing session: <https://youtu.be/gd-aCsPKVPo?si=08vqAks8rS5FOUTU>

6

Participants feedbacks and Mentimeter Analysis



Participants emphasized the importance of the discussions in platforms like Nepal Shelter Forum as a key platform for coordinating and exchanging insights on housing reconstruction and recovery programs, specifically those related to the lesson learnings from the 2015 Nepal earthquake. Participants noted that the forum went beyond typical discussions on disaster risk reduction and shelter reconstruction, recovery. The involvement of experts and specialists, including high-ranking government officials (Ministers, Secretary and Joint Secretaries), Former Ministers, Former Chiefs, Members of the NPC, was seen as crucial in collecting strategic inputs, promoting partnerships

and informed decision-making interventions. Participants shared valuable feedback pertaining to sessions that discussed mainly on - i) Measures for fire risk and preparedness; ii) Approaches that cut across various sectors in reconstruction, such as GEDSI, GRID, ICT; iii) Integrated approaches to reconstruction and recovery and, iv) Multi-hazard disaster management approaches.

Participants stressed the need for coordination among different actors and sectors, emphasizing the continuation of ongoing efforts to localize interventions. The NSF Organizing Committee particularly expressed great satisfaction in

bringing together leaders from across the country, showcasing the partnership between the Nepal government and development partners. There was a shared belief that these collaborations aimed to promote sustainability and inclusivity in the region by advocating for environmentally friendly

and locally sourced building technologies. Overall, participants expressed gratitude to the organizers for providing a platform to share experiences and connect with inputs of experts and specialists that are dedicated to building a more resilient infrastructure.

Are there any broad thematic areas you would like to be included in the next NSF?



Promotion of local building practices

Native or local Social support practices on house building.

Innovative inclusive technologies for resilience building

Intersectuality and social protection in DRR...

Structural n non structural approaches for landslide protection

Women empowerment

Case studies of context and climate suitable houses and settlements, showing routes forward

Climate and grid more focus

Are there any broad thematic areas that you felt we needed to be included but was not included?



I think a separate session should have been done about why we failed on implementing local knowledge and skills in reconstruction

No

Disaster Risk Financing in terms of ShelterResilient Inclusive and Sustainable Construction Technologies

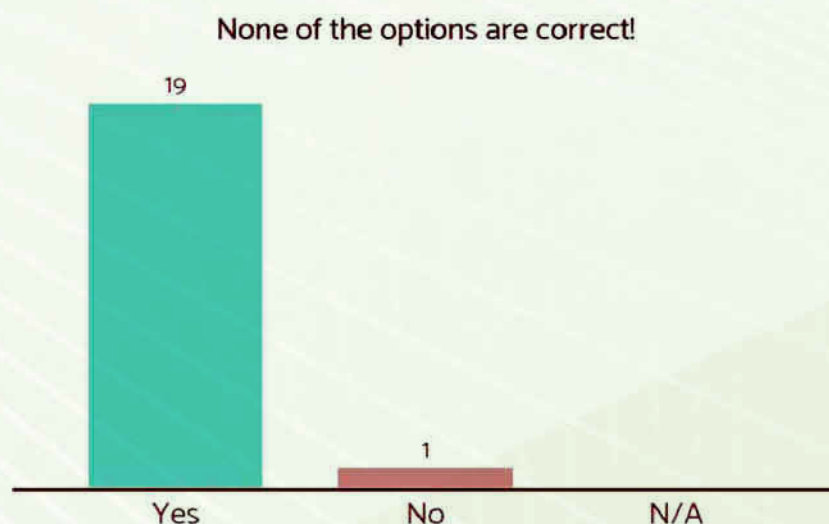
यो दुई दिने छलफल ज्यादा महत्वपूर्ण थियो । यो छलफलमा १० प्रकारका अपाङ्गतालाई पनि सहभागी गराउनु भएको भए अझै राम्रो हुने थियो ।

Data Security

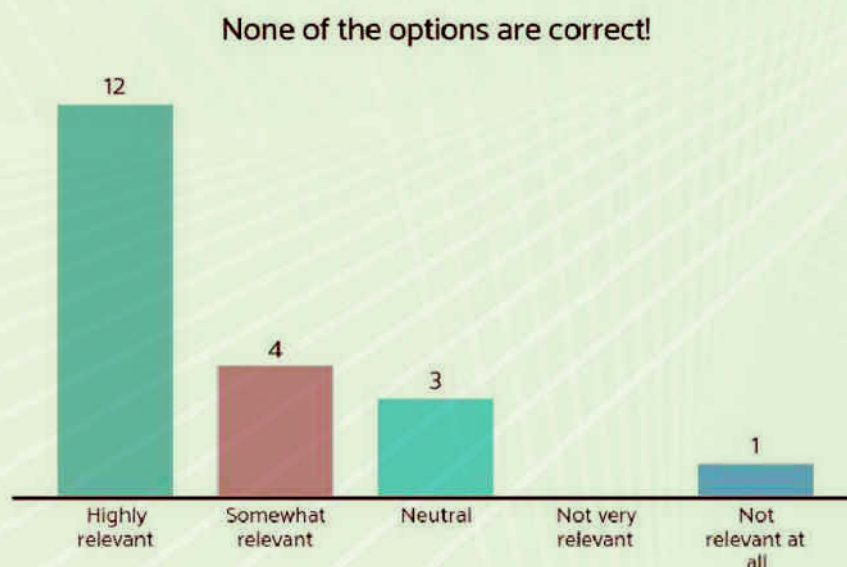
Practical resilient shelter solutions- technology and institutional set up

The role of NGO, INGO and government in post earthquake recovery

Did you find the networking opportunities provided during the event valuable for exchanging ideas and building connections?



Were the topics covered in the forum relevant and aligned with the current challenges and needs in the shelter sector?



Are there any broad thematic areas you would like to be included in the next NSF?



Promotion of local building practices	Native or local Social support practices on house building.	Innovative inclusive technologies for resilience building	Intersectinality and social protection in DRR...
Structural n non structural approaches for landslide protection	Women empowerment	Case studies of context and climate suitable houses and settlements, showing routes forward	Climate and grid more focus

Do you have any feedback or suggestions for organizers of the NSF 2024?



Need pre timeline for preparation, need differnciate of themes	Shorten the remarks from Chair of panel sessions	Facilitators were speaking more than panelists in some of the sessions	These issues should be reached out to PM so next time ensure PM's presence
Take care of time	Time Management	Time consuming lunch can be packed to the hall.	Organize 1 day session of similar subjective theme than putting everything in lumpsum

Are there any broad thematic areas that you felt we needed to be included but was not included?



I think a separate session should have been done about why we failed on implementing local knowledge and skills in reconstruction	No	Disaster Risk Financing in terms of ShelterResilient Inclusive and Sustainable Construction Technologies	यो दुई दिने छलफल ज्यादा महत्वपूर्ण थियो । यो छलफलमा १० प्रकारका अपाङ्गतालाई पनि सहभागी गराउनु भएको भए अझै राम्रो हुने थियो ।
Data Security	Practical resilient shelter solutions– technology and institutional set up	The role of NGO, INGO and government in post earthquake recovery	

Are there any broad thematic areas that you felt we needed to be included but was not included?

No how far i got... it was almost covered	Employment	Multi hazard Risk sensitive land use planning.	Affordable housing
Issues of landless communities in terms of reconstruction and recovery Partnership with academic institutes	Electrification of cooking	Involvement of academic sector for disaster preparedness and recovery	Involvement of filed experience officers

Are there any broad thematic areas that you felt we needed to be included but was not included?

No	Yes	Interlink between researcher and general public	Intersectionality and social protection iin DRR
Real working scenario in the field	Empirical research papers from researcher and academic institutions	GRID	Landslide mitigation, protection measures linking with forestation

What is your one major takeaway from NSF 2024?

partnership, collaboration among stakeholders	Non duplication of similar activities/policies	Resilient Shelther	Research, Collaboration and investment are necessary
Housing and shelter construction is integrated and multi-sectoral initiatives	Interlink of Government, NGO AND INGO in post earthquake recovery	GEDSI Localization...	Important measures , near by policies , disasters and kowledge abt the hazards of climate chage..

7

Key Recommendation and Way Forward

Implement standardized methods for local data collection to ensure consistency and accuracy across disaster management and reconstruction efforts.

Ensuring equitable access to ICT tools for disaster risk management is crucial, underscoring the importance of addressing the digital divide to maximize the effectiveness of ICTs.

Effective advocacy and disaster management require strong local and donor partnerships, focusing on placing communities and individuals at the center of the implementation process.

Utilize advanced technologies for resilient disaster management and reconstruction, including GIS, remote sensing, and data analytics.

Ensure that the needs of vulnerable populations are prioritized in reconstruction and recovery efforts, with a focus on inclusivity and equity.

Emphasize the “build back better and safer” agenda to enhance resilience and safety in rebuilding efforts, considering the specific needs of marginalized communities.

Facilitate collaboration between federal, provincial, and local governments on shelter reconstruction and rehabilitation for earthquake and multi-hazard affected communities.

Deploy community volunteers effectively in disaster management, particularly in fire response & mitigation and beyond, to enhance local response capabilities and community resilience.

Integrate indigenous knowledge, inclusivity, sustainability, and resilience into shelter policies and reconstruction efforts.

Engage the academic sector in disaster preparedness and recovery to integrate science-based learning and innovative solutions.

Ensure that anticipatory actions include an

intersectional GEDSI perspective to promote inclusivity.

Encourage the use of environmentally friendly, nature-based solutions and locally sourced building materials, aligned with the National Building Code, for multi-hazard resilient shelters.

Rampantly conduct local awareness programs and preventive measures against fires, enhance firefighting capabilities, and conduct research on air pollution related to fires.

Explore the development of flood-resilient shelters using the locally available nature-based solutions and preparedness codes.

Implement a “no one is left behind” approach, ensuring that minor disasters are given equal importance.

Prioritize livelihood opportunities in post-disaster response and recovery alongside shelter reconstruction.

Hold periodic discussion forums at provincial, regional, and local levels to boost ownership and promote localized, holistic approaches.



Annexes

Annex1: Questions and Answers:

Q. What are the current discussions or work out being done in the government policy for integrated approach as in the earlier 2015 earthquake not all sectors could be addressed like shelter reconstruction?

A. In 2015, PDNA attempted to provide an integrated package to the earthquake victims. While implementing, the urgency for housing was felt acute and which drew national and international attention. It brought housing to the forefront. In addition, other agencies such as Water Supply, irrigation, agricultural and Industrial agencies provided some elements of the integrated packages. At present, legal and institutional set up has changed drastically which has brought the local governments at the center. They are designated as the delivery agency. The central issue here is their capacity - both institutional and financial. All other stakeholders: federal, provincial, donors and civil society members have to engage with local governments to capacitate them to deliver the integrated reconstruction and recovery interventions. It may have some additional gestation period for realization.

Q. How can we consider the inclusion of women and marginalized communities in the reconstruction and recovery?

A. Some proven way forward for inclusion of women and marginalized groups in reconstruction and recovery are- i) mandatory provision to have ownership of houses in the name of husband and wife which has been used in several recovery project such as Gujarat 2001 earthquake, Tsunami 2004 in Indonesia and in Nepal 2015 also; ii) certain percentage of women and marginalized group need to be trained in skill developments programs such as mason training mainly; iii) STA access equally for recovery of women-headed households and for marginalized groups; iv) a cash for work program to remove debris by targeting

assistance to marginalized and vulnerable people by injecting some cash as supplement to their income and labor, this measure was taken in 2023 November Jajarkot earthquake by many organizations; v) construct disable friendly features while constructing temporary and permanent houses.

Q. What went well regarding the integrated approach of reconstruction and recovery in the 2015 earthquake and what are the learning for the upcoming disaster event?

A. The Community Mobilization Program (CMP) adopted in the earthquake housing reconstruction project was successful. The main feature of CMP was to train and hire local masons as mobile masons who would directly and closely support the beneficiaries. Parallel implementation of reconstruction projects as well as DRR & Risk Analysis projects was a good approach. The projects were linked with each other where output of one project was used in other projects. Better planning and support for mainstreaming is necessary for the output of DRR projects (especially planning, mapping and hazard/risk analysis) to be utilized by the Government in the future activities. Integrating different schemes of support like technical support, grant aid, financial assistance mainly was successful.

Q. What could be the way forward that the DUDBC should follow in order to formulate a comprehensive Building Code Systems or Fire Safety Code?

A. There is a need to update the existing building code to include the latest international fire safety standards and technologies, along with making mandatory & regular fire safety audits for all buildings. Similarly, collaboration among fire departments, urban planners, architects, and the public and local governments and communities at large should be robust. Frequent training and awareness programs should be conducted at public-level from

various communication channels, including strict enforcement of fire safety regulations with penalties for non-compliance at community level. Comprehensive guidelines for emergency preparedness and response are also required at various levels of government and the regular revision of existing code to adapt with the new challenges and advancements in fire safety.

Q. What measures could be implemented to effectively control the forest fires?

- A. Some measures can be effective to mitigate and control the forest fires such as - early detection of forest fires from use of ICT tools like satellite monitoring and ground sensors; create and maintain firebreaks to slow the spread of fires in other areas; increase public awareness about the causes and prevention of forest fires through education campaigns; engage local communities in fire prevention and response efforts; establish and train rapid response teams and volunteers squad equipped to deal with forest fires quickly and effectively; implement and enforce strict regulations regarding activities that could trigger forest fires, such as campfires and the disposal of flammable materials and implement sustainable forest management practices to reduce the accumulation of combustible materials.

Q. What kinds of forthcoming project initiatives are in the planning for Nepal to counteract climate change and its consequences?

- A. There are programs on climate adaptation and resilience for South Asia implemented by the ADPC and RIMES in technical indicator and decision-making support in the transport, agriculture and climate change disaster sector with the collaboration of DHM.

Q. What are the strategic suggestions to the Nepal government to increase emphasis on the GRID in its plans and programs?

- A. The World Bank implemented a greener cooling pathways program in India which has relevance to replication in Nepal too. Along with this, there are possibilities of bringing programs in scaling up with new technologies while preserving the traditional architectural infrastructures with a view of the environmentally friendly approach.

Q. The DWRI/MoEWRI is currently focused on river training works. How can the riverbank map prepared by DWRI/MoEWRI integrate with other departments?

- A. We recently passed the policy on water-induced disaster management. As part of our advocacy, we are working on developing the corresponding act and the river management plan. This plan will specify the areas designated for natural water flow and sediment accumulation. The current policy on water-induced disaster management clearly defines the setback distance for natural water flow, as well as the areas for sediment and risk zones.

Q. What are some of the works in flood-resilient solutions starting from river management aspects? How are the collaborations working out?

- A. Currently, the DHM has completed risk zone and sediment mapping. We have not yet worked on developing flood codes for building resilient shelters, as we are waiting for specific Acts/ Policies to guide us further. With support from the ADB, we are implementing the Priority River Basin Project, which includes the Community-Based Disaster Risk Reduction and Management (CBDRM) initiative. This project involves constructing 48 shelter houses.

Q. How is the GEDSI approach integrated and progressing in the UN's programs and beyond, particularly in disaster management?

- A. Integration of the GEDSI approach at all levels of government and stakeholders' initiatives is crucial. Programs should be designed with a gender perspective, from relief efforts to response and recovery. All stakeholders must invest in and implement strategies to integrate the GEDSI approach effectively. For successful disaster management, strong data management is essential. However, there has been a notable lack of data management at the local level, resulting in insufficient information to identify vulnerable communities and groups. To address this, data collection must be robust and comprehensive. Additionally, mobilizing local community volunteers, especially female volunteers, has proven to be an effective strategy in integrating the GEDSI approach in

disaster management and addressing the needs of families with female members.

Q. What are the delays in distributing the second installment for temporary shelter construction?

A. The delays are primarily due to the ongoing process of finalizing the PDNA report. Additionally, issues with updating data in the Bipad portal and unclear database information, along with the need for data verification at the local level, complicate the situation. Accurate information dissemination at all levels is essential to expedite response efforts and support disbursement.

Q. What are the ways forward for multi-hazard disaster management in Nepal and the relevant response, recovery, and reconstruction efforts?

A. Nepal's vulnerability to various natural disasters necessitates a comprehensive approach to multi-hazard disaster management. Strategies must be robust and adaptable to different hazards, such as earthquakes, floods, and landslides. A significant concern is the lack

of adherence to building codes, which makes structures highly vulnerable. Strict enforcement of building code regulations is urgently needed, and local governments must seriously endorse and implement these measures. All new constructions must comply with building code standards, with no exceptions.

Q. What are the roles and responsibilities in disaster management at the local level, and how is the budget allocated across various titles?

A. Roles and responsibilities at the local level are clearly defined, with budget allocation mapping conducted from the ward level, where plans are prioritized based on necessity. However, issues such as misinterpretations and delays in service delivery due to the deployment of fresh and unexperienced technical manpower persist, hampering field-level implementation. Resource financing and staff allocation remain significant challenges at the local level.



Annex2: Market Stall and Networking





Annex2: Registration and Networking





Compressed Agenda

Day I, 7 May 2024 9:00 am to 5:00 pm		From Recovery to Resilience: Localizing National Policies and Frameworks for Multi Hazard Resilience		
Lead Agency		National Disaster Risk Reduction and Management Authority (NDRRMA)		
Supporting Organization		WB/MDTF, UNDP, HFHN, Build UP, NRCS/IFRC, Save the Children, BEEN, Caritas Nepal, SY Panel, CRS, WVI, IOM Member Secretariat: National Housing and Settlement Resilience Platform (NHSRP)		
Time	Duration	Session Topic	Session	Facilitator
8:00 - 9:00	60 min	Breakfast and Registration		
9:00-10:30	90 min	Introduction by Ms. Ranjana Shrestha, Master of Ceremony	Opening Session	Ms. Ranjana Shrestha, Master of Ceremony
		Security Briefing by Mr. Rahul Pratap Singh, Admin and Finance Coordinator, NHSRP		
		Welcome remarks by Mr. Rameshwar Marhatta, Joint Secretary, NDRRMA		
		National Anthem		
		Moment of Silence		
		Inauguration: Lighting panas by the Chief Guest Honourable Minister Mr. Bhanubhakta Joshi, Ministry of Federal Affairs and General Administration (MoFAGA)		
		Remarks by NSF Secretariat Ar. Reshma Shrestha, Deputy National Coordinator, NHSRP		
		Message from the Philippines DHSUD Organizer for the Asia Pacific Shelter and Settlements Forum (APSSF) 2024		
		Remarks, Honourable Minister Ministry for Urban Development Mr. Yamlal Kandel, Chief Minister, Karnali Province		
		Remarks by Chief Guest: Honourable Minister Mr. Bhanubhakta Joshi, MINISTRY OF FEDERAL AFFAIRS & GENERAL ADMINISTRATION		
		Closing of Opening Session Remarks: Mr. Anil Pokhrel, Chief Executive, NDRRMA		
10:30 - 10:45	15 min	Discussion with participants over Tea (Market stall Visit)		

10:45 - 11:45	60 min	Panel: Multi-hazard Recovery and Reconstruction Governance Facilitator: Mr. Jagdish Kharel, Sr. Journalist Panelist: 1. Mr. Yamlal Kandel, Chief Minister, Karnali Province 2. Ms. Ramkumari Jhakri, Former Minister, Ministry of Urban Development 3. Mr. Sushil Gyewali, Former Chief Executive, National Reconstruction Authority 4. Mr. Anil Pokhrel, Chief Executive, NDRRMA		
11:45-12:30	45 min	Presentation: Reflection of West Nepal Earthquake 1. Development Partner Presentation	Technical Session	
1:30 - 02:30	60 min	Discussion with participants over Lunch (Market stall visit)		
2:30 - 3:30	60 min	Panel: Fire Risk and Preparedness Measures Facilitator: Mr. Bhushan Tuladhar, Chief of Party, FHI360 Panelist: 1. Prof. Dr. Rejina Maskey Byanju, Professor, Central Department of Environmental Science (TU) 2. Mr. Baijnath Prasad Yadav, Mayor, Madhav Narayan Municipality, Rautahat 3. Mr. Sundar Prasad Sharma, Under Secretary, NDRRMA 4. Ms. Parbata Gautam, General Secretary, FECOFUN 5. Mr. Ranjan Dhungel, Director, NSET	Panel Discussion	Chair: Er. Machakaji Maharjan, Deputy Director General, DUDBC
03:30-04:30	60 min	Panel: Enhancing Multi-Hazard Disaster Risk Reduction Approaches and Technologies for Resilient Shelter Facilitator: Dr. Ramesh Guragain, Deputy Executive Director, NSET Panelist: 1. Dr. Archana Shrestha, Deputy Director General, DHM 2. Ms. Monika Jha, Deputy Director General, DMG 3. Mr. Prem Lasiwa, Senior Divisional Engineer, DWRI/MoEWRI 4. Mr. Basanta Paudel, Disaster Risk Management Specialist, World Bank	Panel Discussion	Chair: Mr. Rameshwor Marhatta, Joint Secretary, NDRRMA
4:30-4:45	15 min	Wrap up for the day and closing remarks: Mr. Rameshwor Marhatta, Joint Secretary, NDRRMA	Day I closing	
4:45 onwards		Discussion with participants over High Tea (Market stall Visit)		

Day 2, 8 May 2024 9:00 am to 5:00 pm		From Recovery to Resilience: Localizing National Policies and Frameworks for Multi Hazard Resilience		
Lead Agency		National Disaster Risk Reduction and Management Authority (NDRRMA)		
Supporting Organization		WB/MDTF, UNDP, HFHN, Build UP, NRCS/IFRC, Save the Children, BEEN, Caritas Nepal, SY Panel, CRS, WVI, IOM Member Secretariat: National Housing and Settlement Resilience Platform (NHSRP)		
Time	Duration	Session Topic	Session	Facilitator
8:00-9:00	60 min	Breakfast and Registration		
9:00-9:15	15 min	First Day Overview and Highlights by Mr. Rijan Gajurel, National Technical Coordinator, NHSRP	Technical Session	
9:15-10:15	60 min	Panel: Integrated Approaches to Reconstruction and Recovery Facilitator: Ms. Radha Verma, Sr. Partnership Officer, UNOPS Panelist: 1. Mr. Chandra Bahadur Shrestha, Former Executive Committee Member, NRA 2. Mr. Sudhir Kumar, Risk Management Specialist, UNDP 3. Mr. Bjorn Sodernberg, Director, Buildup Nepal 4. Mr. Gaurav Panthi, Program Officer, JICA	Panel Discussion	Chair: Mr. Anil Pokhrel, Chief Executive, NDRRMA
10:15-11:15	60 min	Panel: Leveraging ICT Engagement for Disaster Preparedness, Response and Recovery Facilitator: Mr. Sachin Malego, National Information Management Officer, NHSRP Panelist: 1. Dr. Iguniwari T Ekeu-wei, Disaster Risk Management Specialist, SSACD, World Bank 2. Dr. Uttam Paudel, Project Director, CRS 3. Mr. Buddhi Sagar Shrestha, Director - DM Department, IFRC 4. Ms. Puja Shakya, Nepal Country Program Lead, RIMES	Panel Discussion	Chair: Mr. Krishna Hari Banskota, Former Chief Information Commissioner
11:15-11:30	15 min	Discussion with participants over Tea (Market stall Visit)		
11:30-12:30	60 min	Panel: Fostering Inclusive Resilience by integrating GEDSI Facilitator: Mr. Gajendra Budathoki, Journalist Panelist: 1. Ms. Reena Chaudhary, GEDSI focal point, NDRRMA 2. Ms. Sama Shrestha, Programme Specialist, UNwomen 3. Ms. Jeevika Khadka, Program Manager -DRRM, Youth Innovation Lab 4. Mr. Sugam Bhattarai, General Secretary, NFDN	Panel Discussion	Chair: Ms. Abha Shrestha Karna, Joint Secretary, MoWCSC.
12:30-01:30	60 min	Discussion with participants over Lunch (Market stall visit)		

01:30 - 02:30	60 min	Panel: Understanding Loss and Damage from Climate Change in the Housing and Settlement Sector Facilitator: Dr. Bimal Raj Regmi, DRR and Climate expert Panelist: 1. Mr. Bijay Prakash Wanem, Chairperson, Maiwakhola Rural Municipality 2. Er. Narayan Prasad Bhandari, Deputy Director General, DUDBC/MoUD 3. Dr. Maheshwar Dhakal, Joint Secretary, MoFE 4. Dr. Bimala Rai Paudyal, Former Minister of Foreign Affairs	Panel Discussion	Chair: Hon. Dr. Arnico Panday, Member of NPC
2:30 - 2:45	15 min	Presentation: Building Energy Efficient Shelter in Nepal, Dr. Daniel Neyer, Project Leader - BUILDING Energy Efficiency in Nepal (BEEN) Project	Presentation	
2:45- 3:45	60 min	Panel: Exploring Innovative GRID Practices for Resilient Shelter, Housing, and Settlements Facilitator: Er. Preeti Karki, GRID and Partnership Technical Coordinator, NHSRP Panelist: 1. Mr. Ganesh Prasad Bhatta, Joint Secretary, MoLMCPA 2. Er. Manoj Nakarmi, Senior Divisional Engineer, DUDBC/MoUD 3. Ms. Eliza Sthapit, Country Director, Habitat for Humanity 4. Mr. Stephen (Steve) Danyo, Program Leader, Sustainable Development, World Bank	Panel Discussion	Chair: Hon. Prof. Dr. Sangeeta Singh, Member of NPC
3:45- 4:00	15 min	Menti- Feedback Collection, Ms. Bina Bagale, MEAL, NHSRP	Technical Session	
4:00- 4:15	15 min	Discussion with participants over High Tea (Market stall Visit)		
		Closing Session		
4:15-5:15		Appreciation and Token of Love Distribution by Chief Guest Closing Remarks Mr. Azmat Ulla, Head of Delegation, IFRC Remarks by Mr. Ek Narayan Aryal , Home Secretary, MoHA Closing Remarks, Honourable Deputy Prime Minister and Home Minister Mr. Rabi Lamichhane Closing Remarks by event Chair, Mr. Anil Pokhrel, Chief Executive, NDRRMA	Event closing	Ms. Ranjana Shrestha, Master of Ceremony
		Group Photo		Mr. Rahul Pratap Singh, NHSRP

2nd National Conference



From Recovery to Resilience: Localizing National Policies and Frameworks for Multi Hazard Resilience

ORGANIZED BY:



Government of Nepal
Ministry of Home Affairs
National Disaster Risk Reduction and Management Authority
Singhadurbar, Kathmandu

SUPPORTED BY

